2026 Fall

CS270: Digital Image Processing

Register here to download lecture slides !

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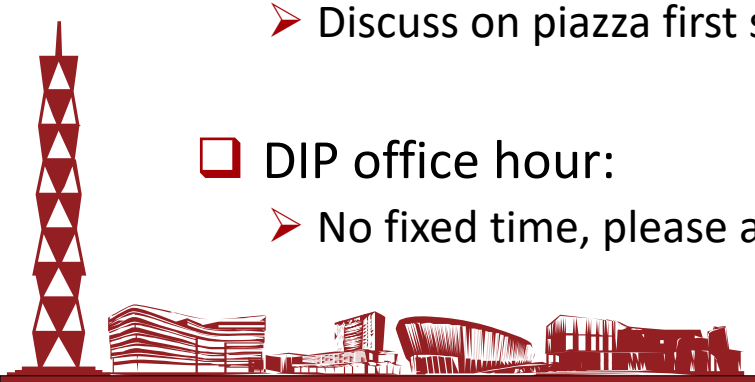
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DIP support team

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- Course discussion system: Piazza
  - https://piazza.com/shanghaitech.edu.cn/fall2026/cs270
  - Please register as fast as possible, content will be shared there
- Course assignments platform: Gradescope
  - https://www.gradescope.com
  - **Details to be announced in Piazza later**
- TAs:
  - WuHao - 伍浩 - wuhao2024@shanghaitech.edu.cn
  - Probably more to come
  - Discuss on piazza first so that others can benefit too
- DIP office hour:
  - No fixed time, please ask on Piazza

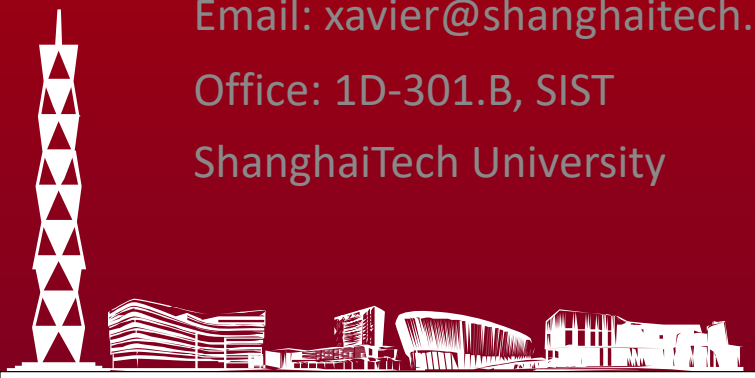


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Lecture 2
Human visual system, perception & pixel

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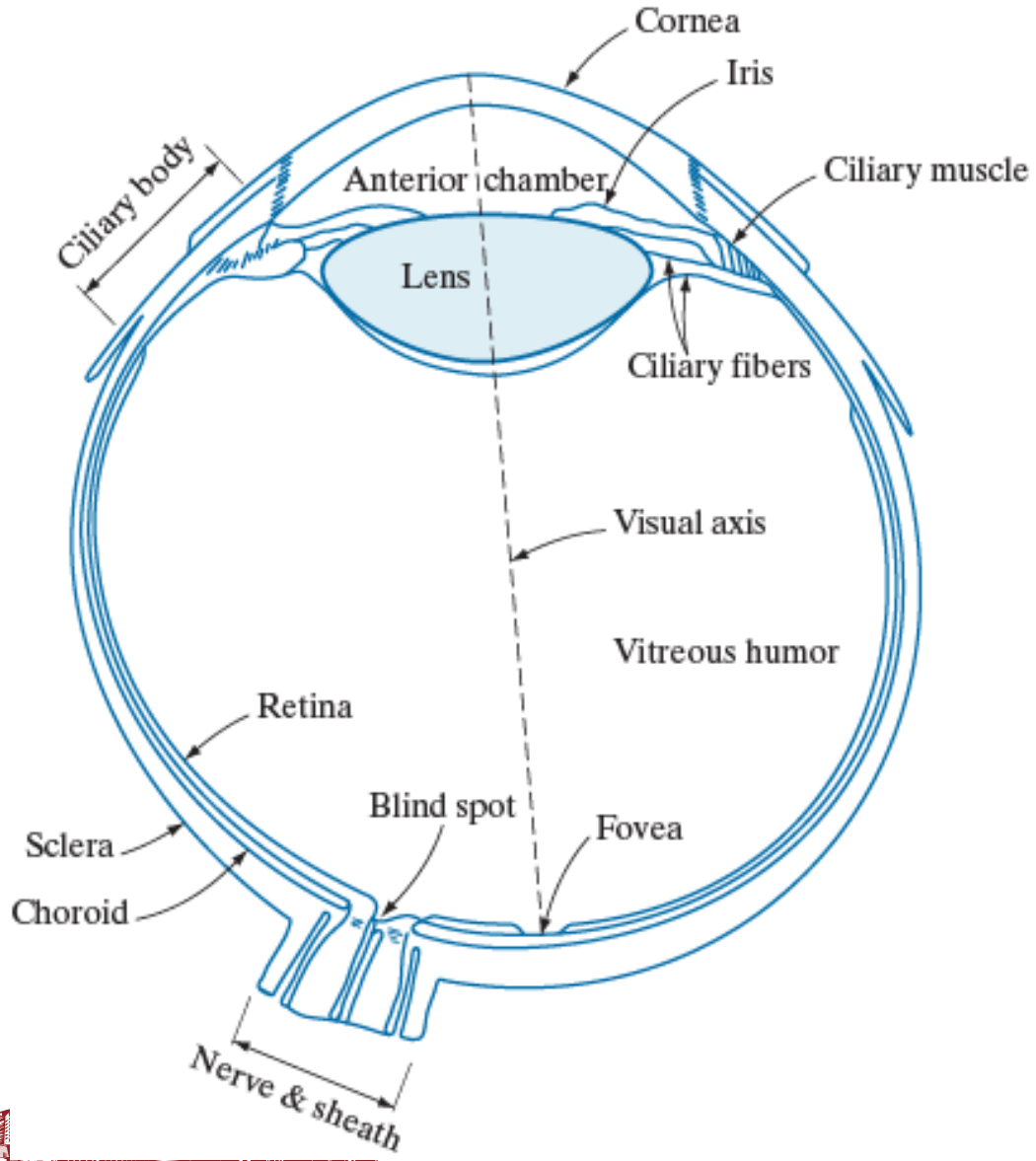
2026 Fall – Sept 20th

立志成才报国裕民

Outline

- Anatomical structure of eye ball
- Image formation and brightness adaptation in the eyes
- Visual illusions
- Pixel
- Basic operation of DIP

Structure of the Human Eye



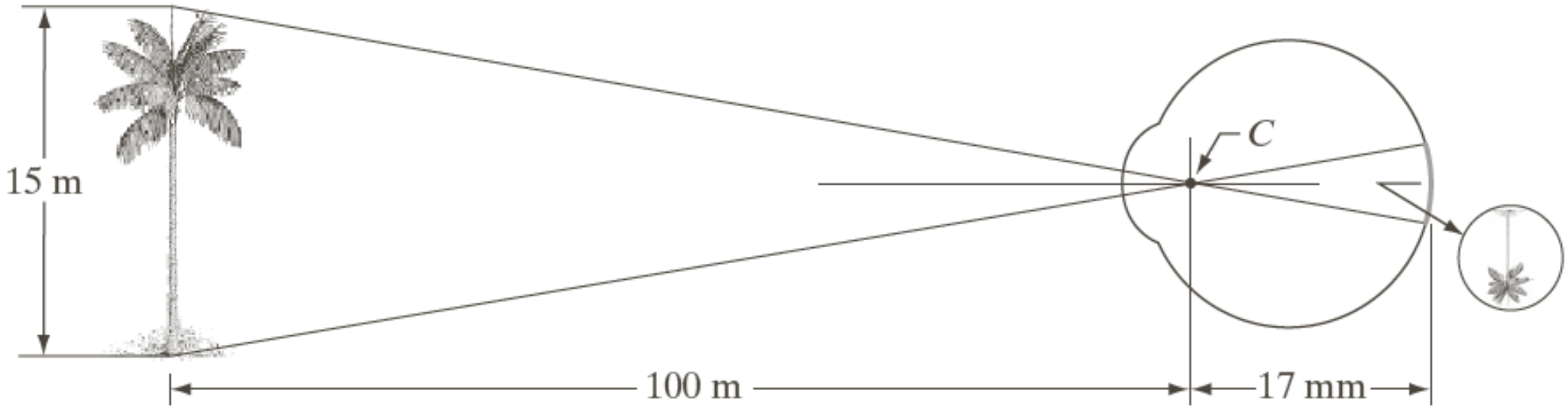
Fovea: 中央凹
Retina: 视网膜
Iris: 虹膜
Pupil: the opening of the iris (2-8 mm) 瞳孔

Lens: 晶状体
Blind spot: 盲点

Receptors:
Rod: 杆状体
(scotopic or dim-light vision)
Cone: 锥状体
(photopic or bright-light vision)

Image formation in the eye

- Objects captured as focused images on the image plane at the retina
  - Perspective projection based on pinhole model geometry,
  - Image size depends on distance of object
  - In practice, optical device with lens



The retina

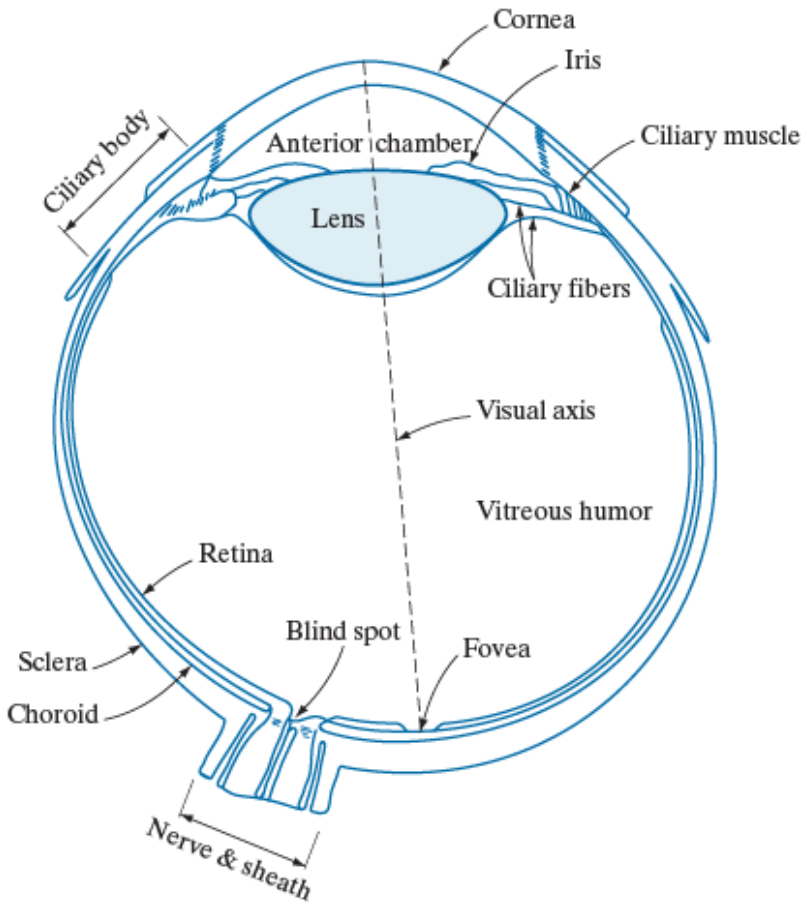


Figure 2.1: Sagittal Eye ball anatomy

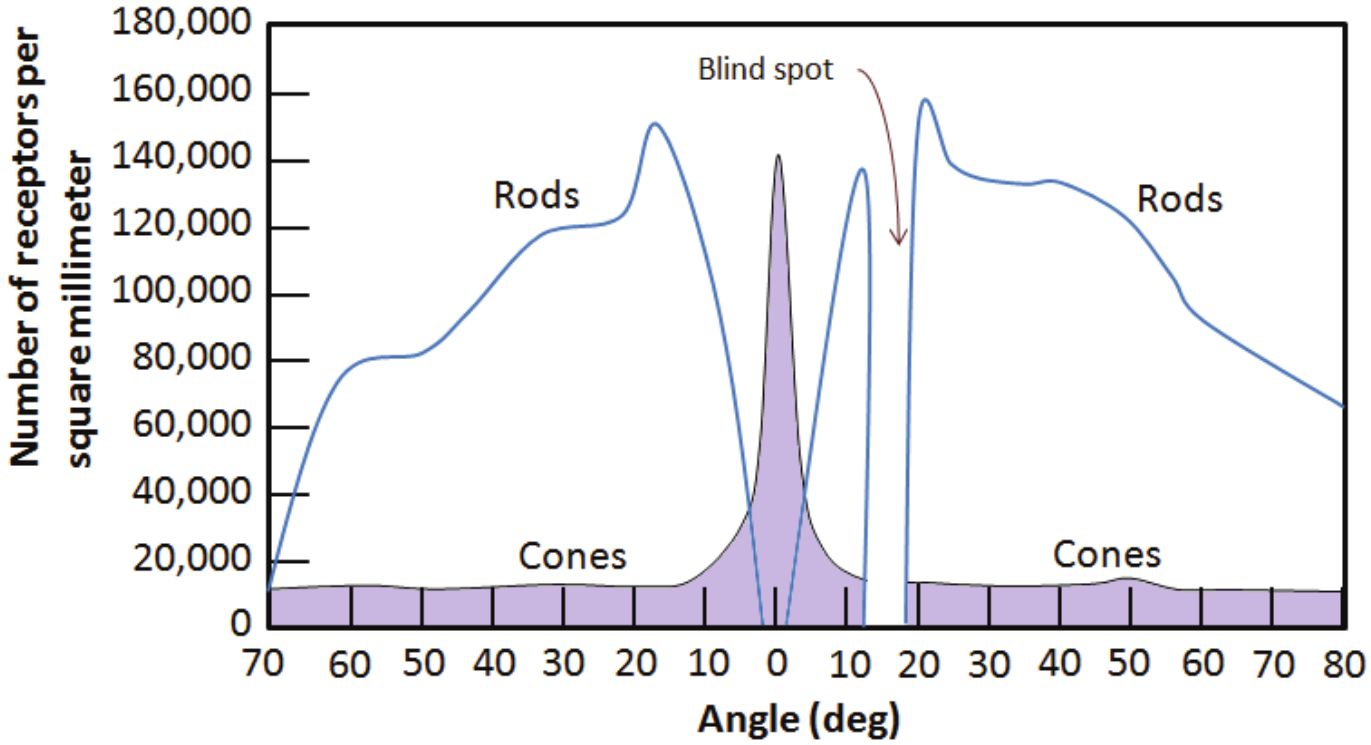


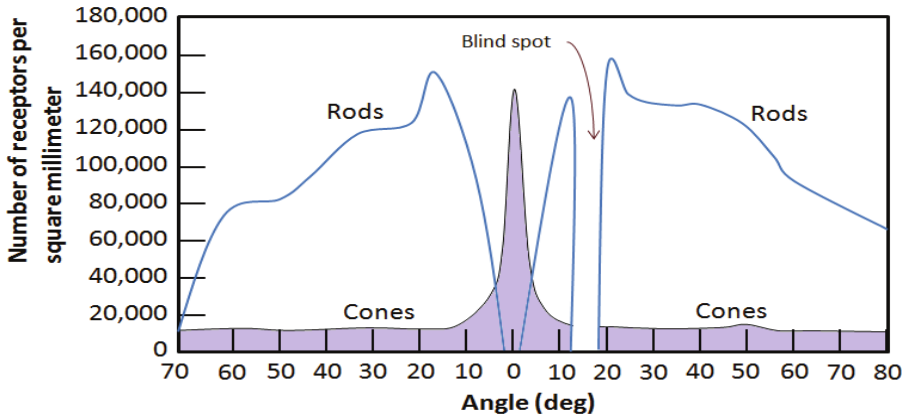
Figure 2.2: Distribution of rods and cones in the retina

Human visual system - perception elements

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❑ Rods:

- Sensitive to light intensity
- Night “scotopic” vision
- Achromatic
- Low acuity (many per nerve end)
- Peripheric vision
- Slow response
- 75-150 Million/Retina



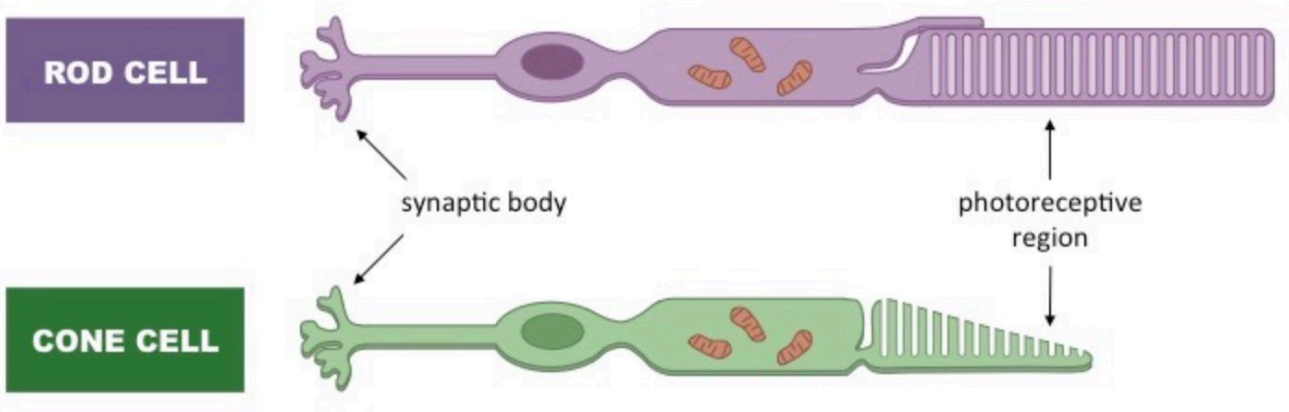
❑ Cones:

- “Photopic” vision
- Chromatic (3 “colors”)
- Concentrated in fovea (1 per nerve end)
- High visual acuity, spatial resolution.
- Fast response
- 6-7 Million/Retina



Rods and cones

Types of Photoreceptors (Rods and Cones)

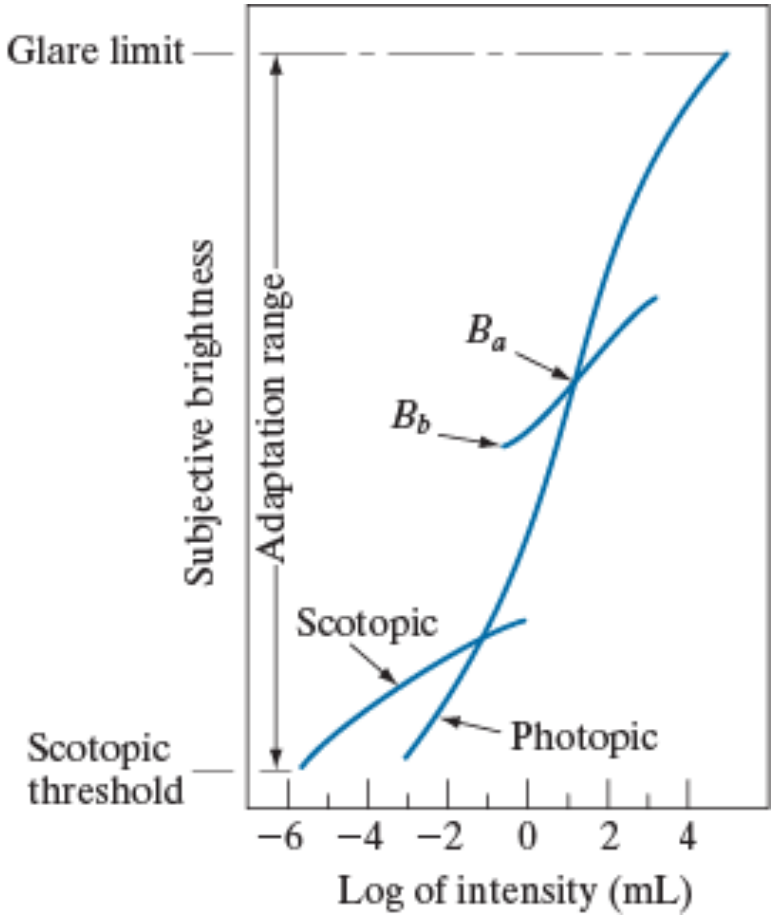


	Rod Cells	Cone Cells
Location in retina	Found around periphery	Found around centre (fovea)
Optimal light conditions	Dim light ('night' vision)	Bright light ('day' vision)
Visual acuity	Low resolution (many rods : one bipolar cell)	High resolution (one cone : one bipolar cell)
Colour sensitivity	All wavelengths	Certain wavelengths (red, green, blue)
Type of vision	Achromatic (black and white)	Colour
Number of types	One (all contain rhodopsin)	Three different iodopsin pigments
Relative abundance	Many	Fewer

Characteristics of Human Visual System

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- **Brightness adaption**
- Simultaneous contrast
- Mach band effect
- Optical illusions



- Impressive total dynamic range ;)
- But, cannot operate over a long range simultaneously ;(
- Brightness adaption level (*Ba*)
- Below *Bb* is **black!**

Characteristics of Human Visual System

- **Brightness adaption**
- Simultaneous contrast
- Mach band effect
- Optical illusions

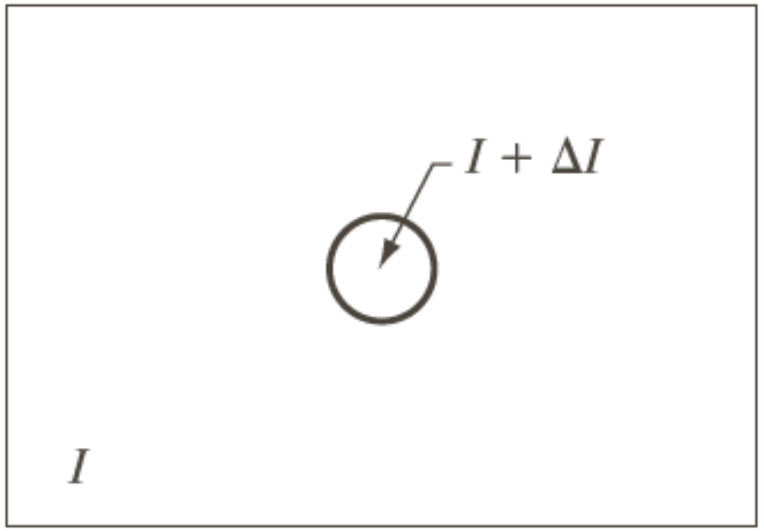


FIGURE 2.5 Basic experimental setup used to characterize brightness discrimination.

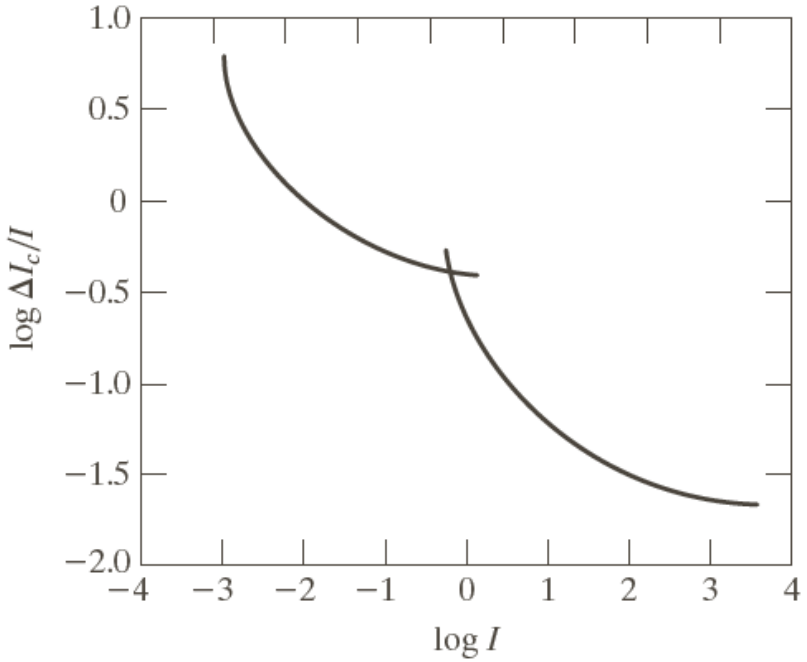


FIGURE 2.6 Typical Weber ratio as a function of intensity.

Characteristics of Human Visual System

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- Brightness adaption
- Simultaneous contrast
- **Mach band effect (Ernst Mach, 1865)**
- Optical illusions

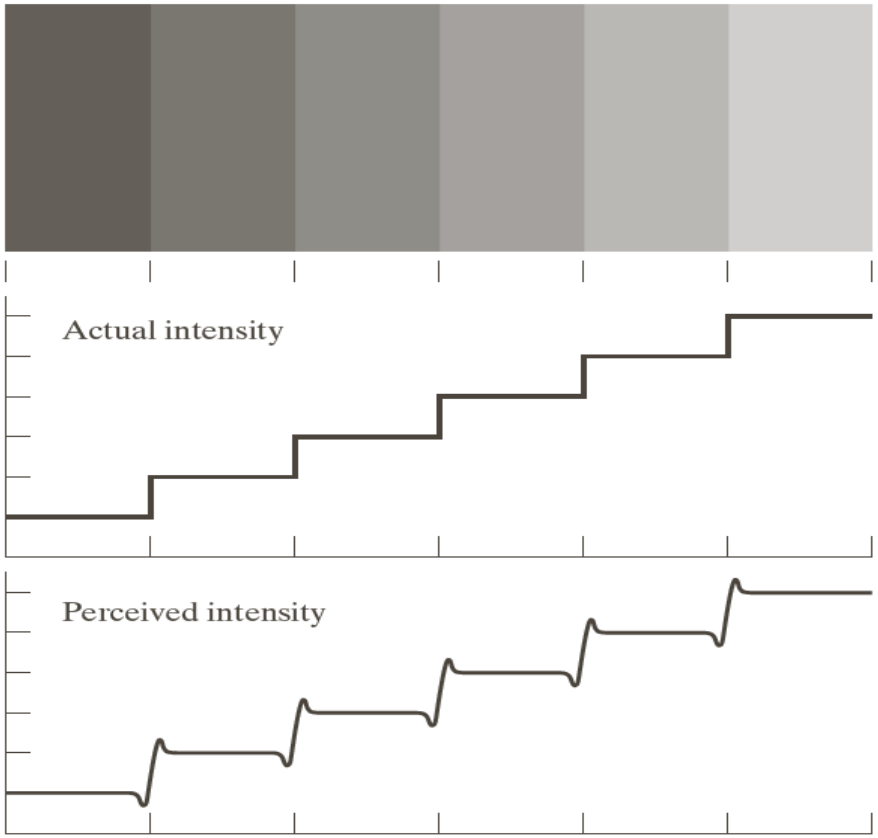
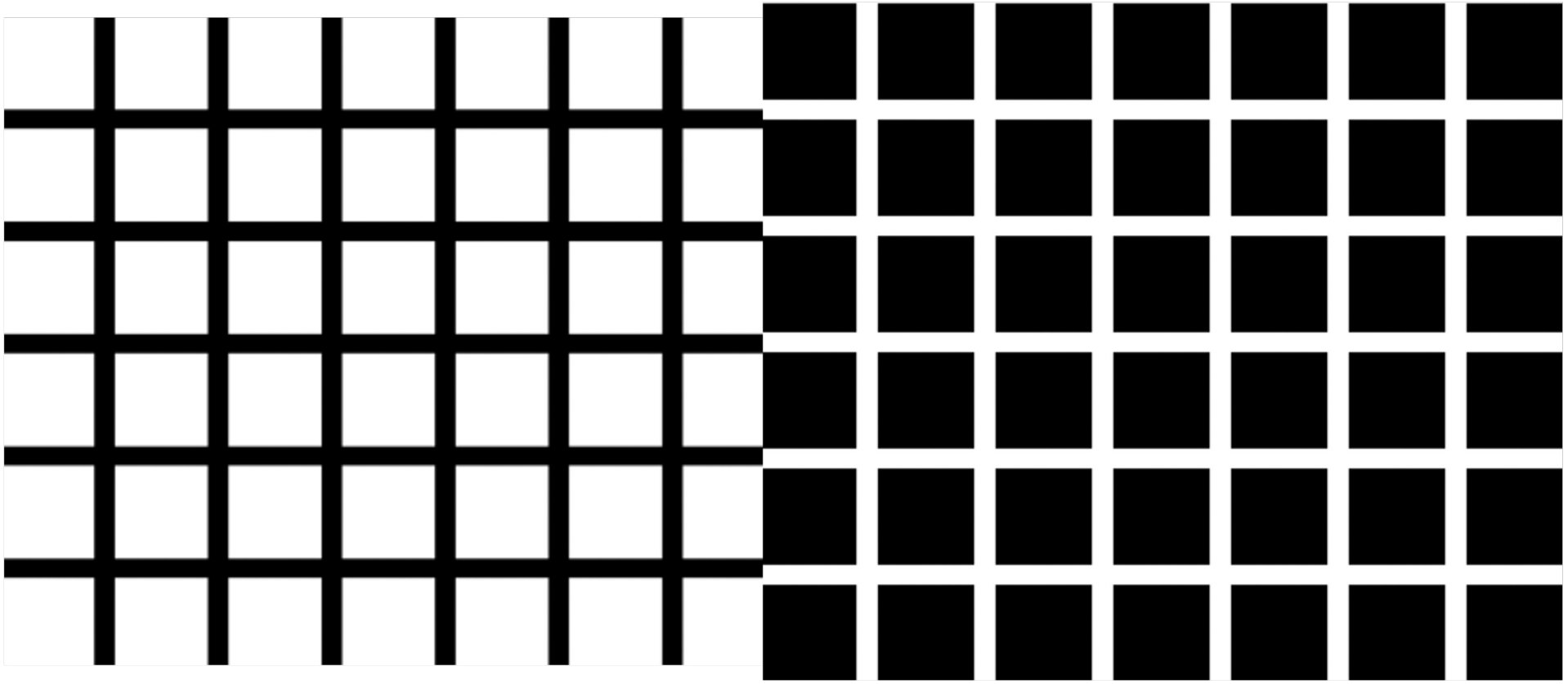


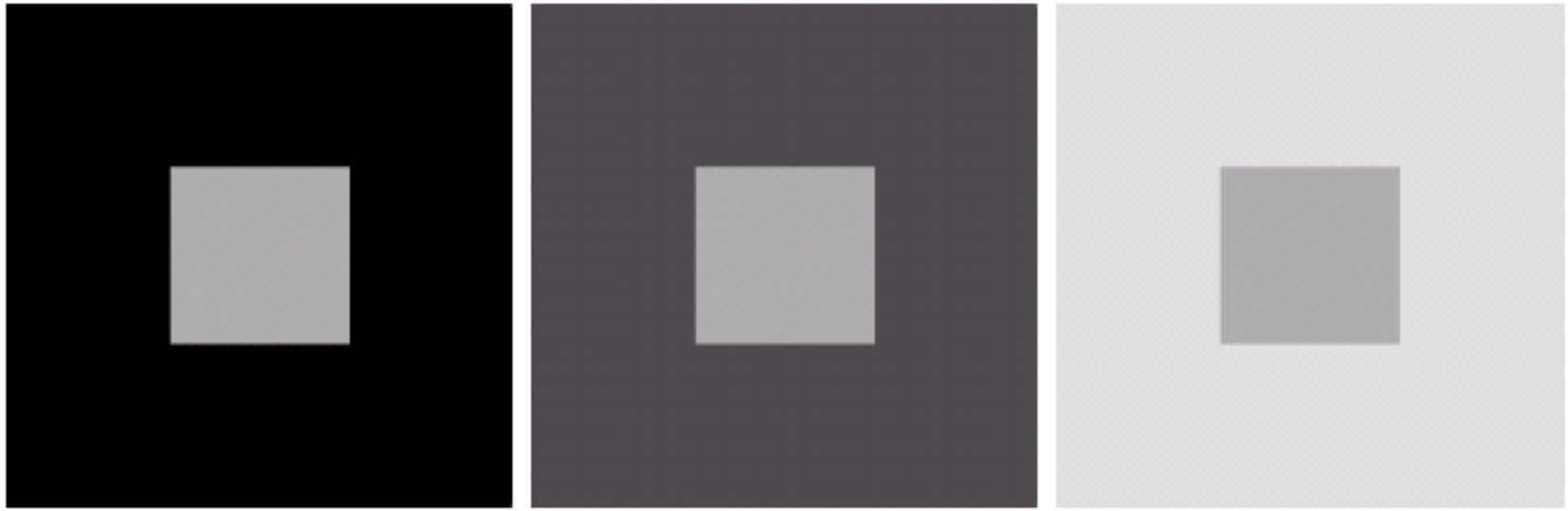
Figure 2.7: Perceived intensity is not a simple function of actual intensity

Characteristics of Human Visual System



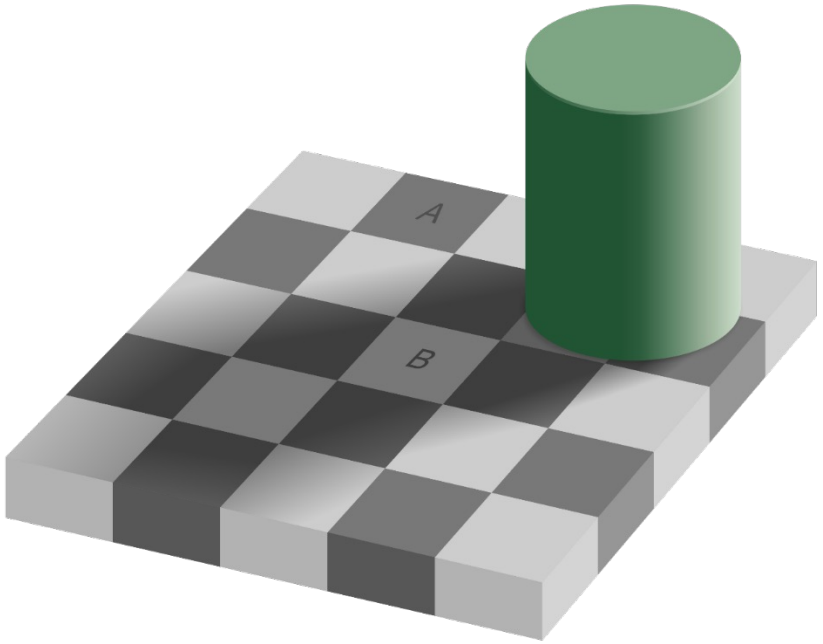
Characteristics of Human Visual System

- Brightness adaption
- **Simultaneous contrast**
- Mach band effect
- Optical illusions



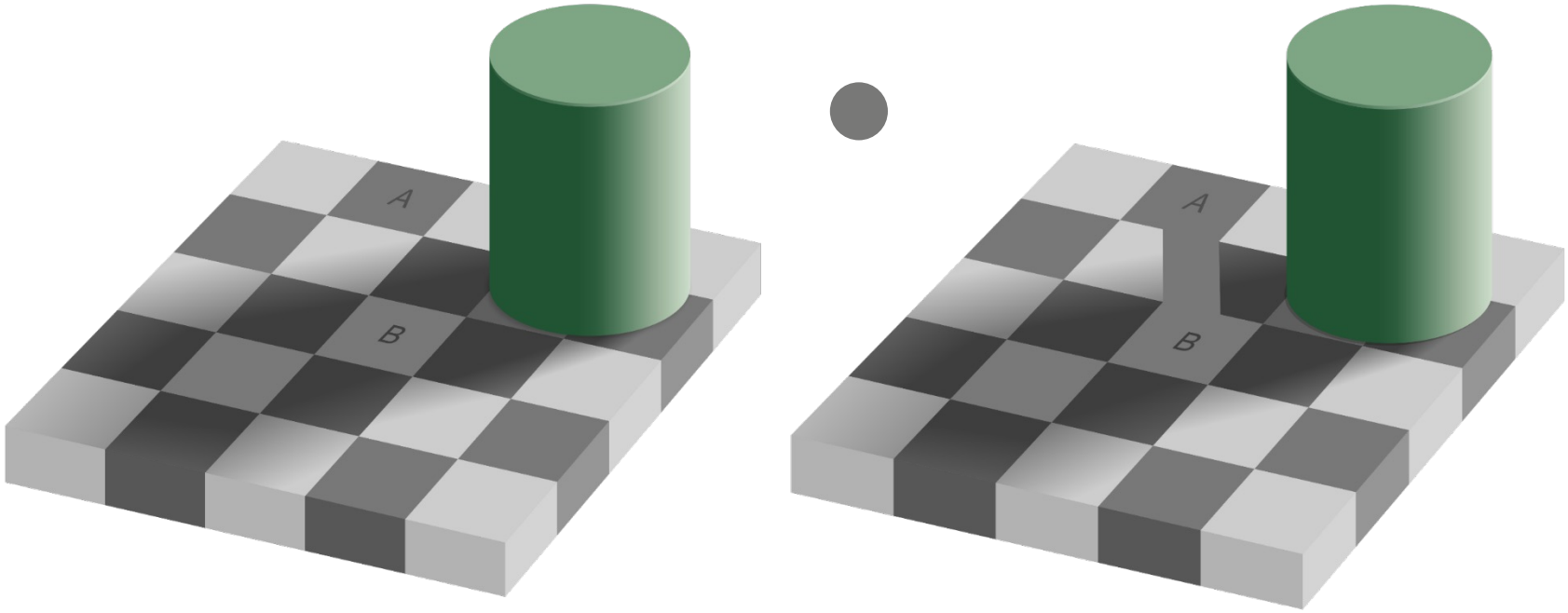
All the inner squares have the same intensity,
but they appear progressively darker as the background becomes lighter

Checker shadow illusion

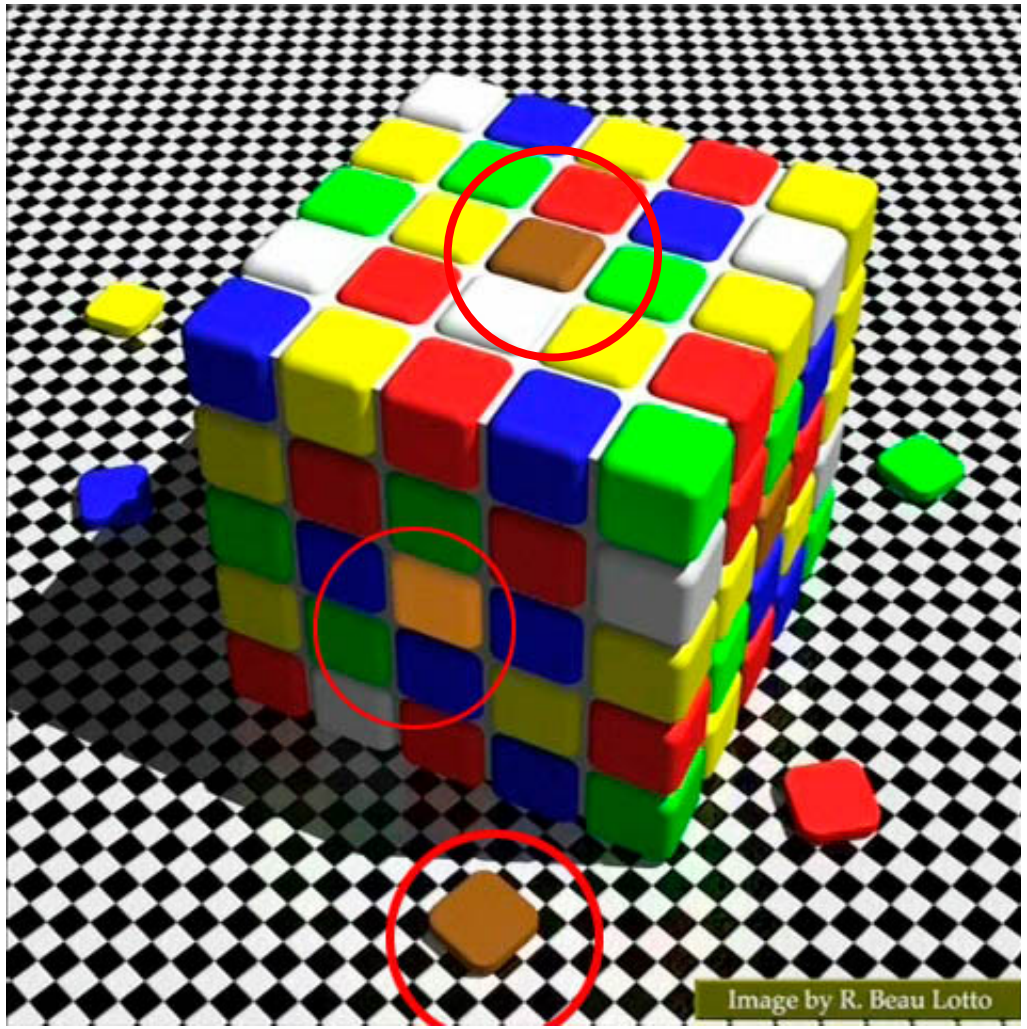


Checker shadow illusion

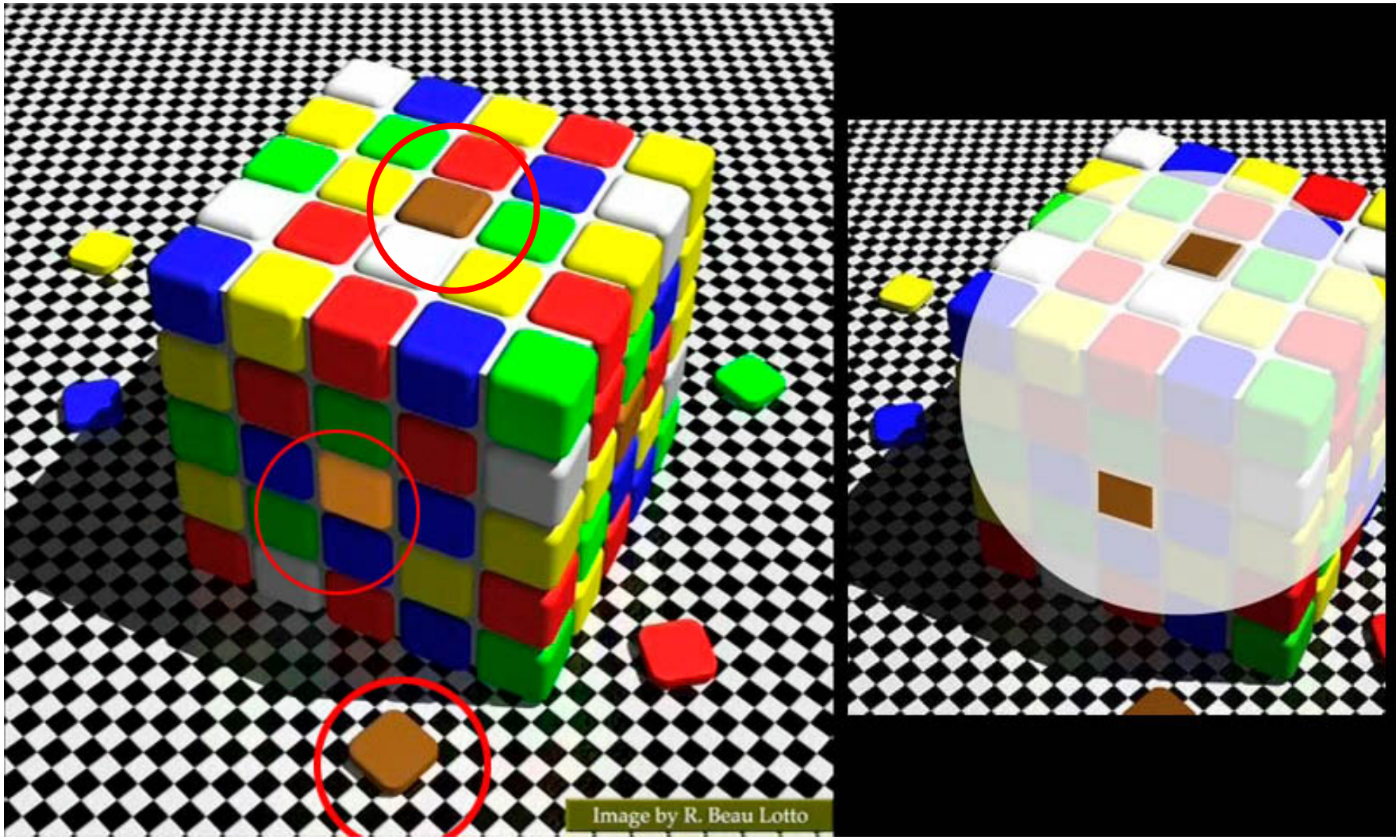
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Cube



Cube

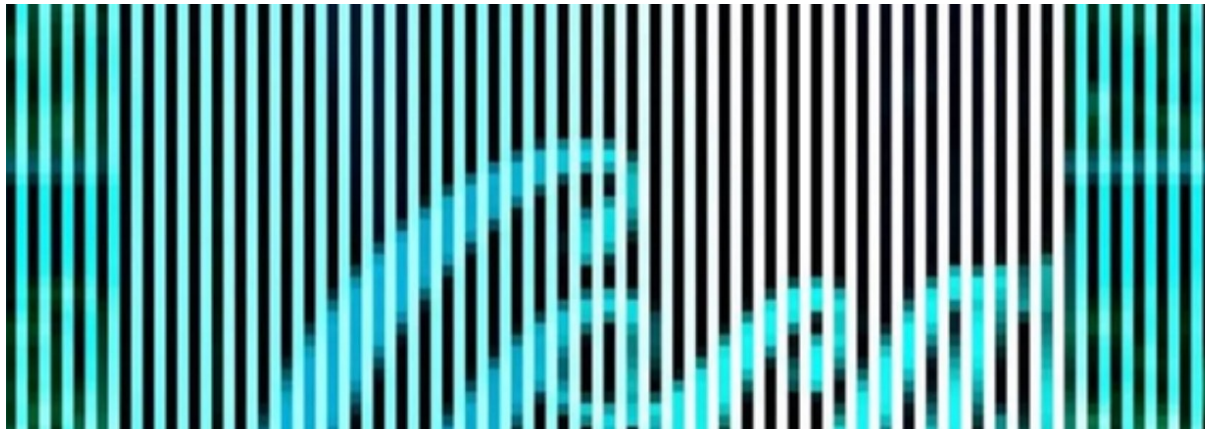


Other color illusions

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Is there some **red** in this picture ?

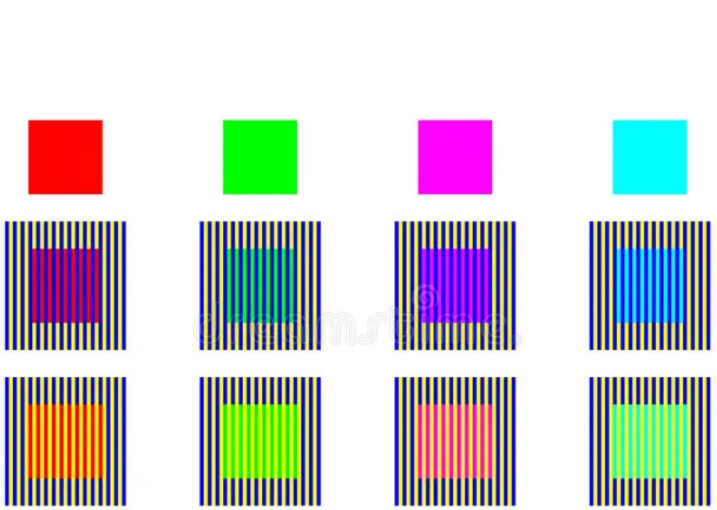


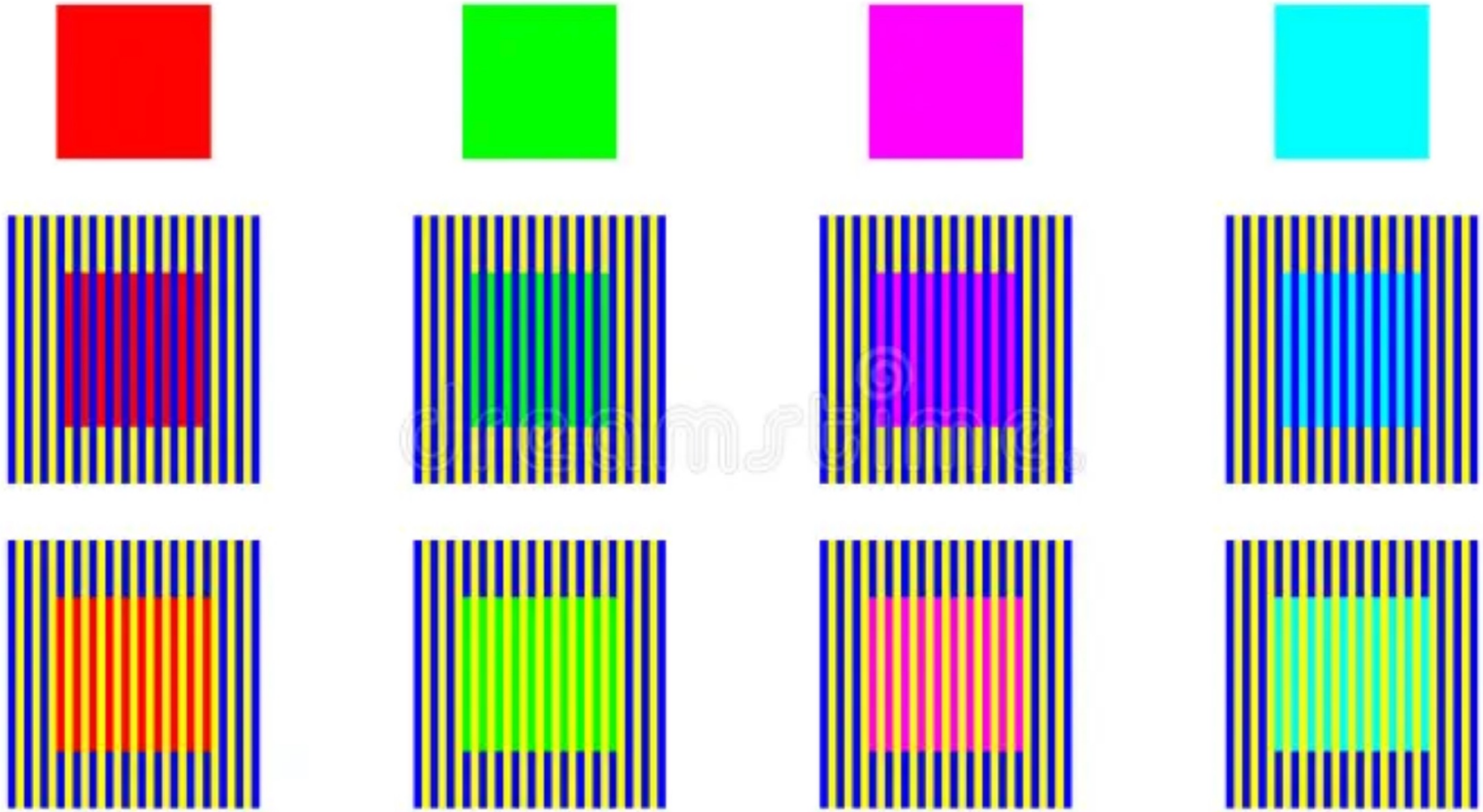




There is no **blue** here either !

Other color illusions

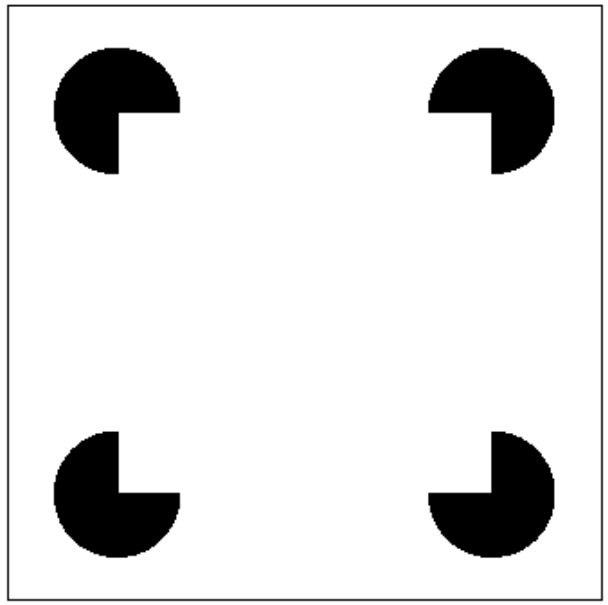


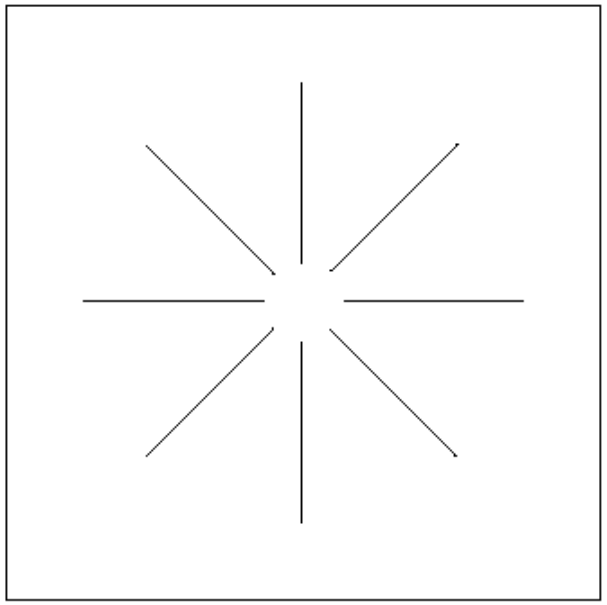


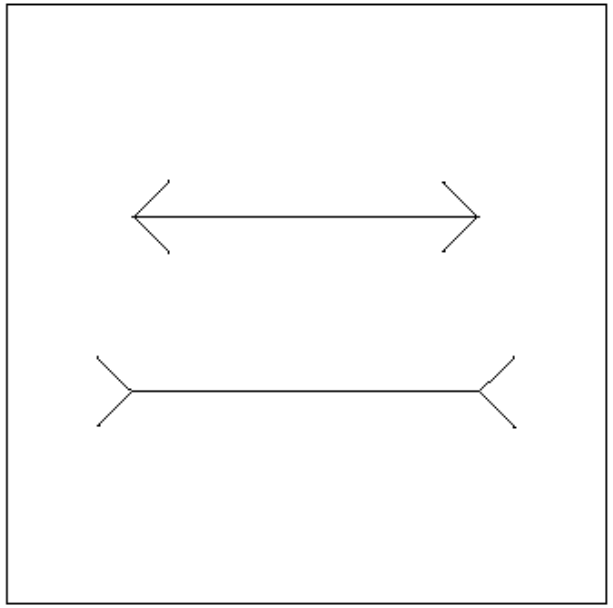
Characteristics of Human Visual System

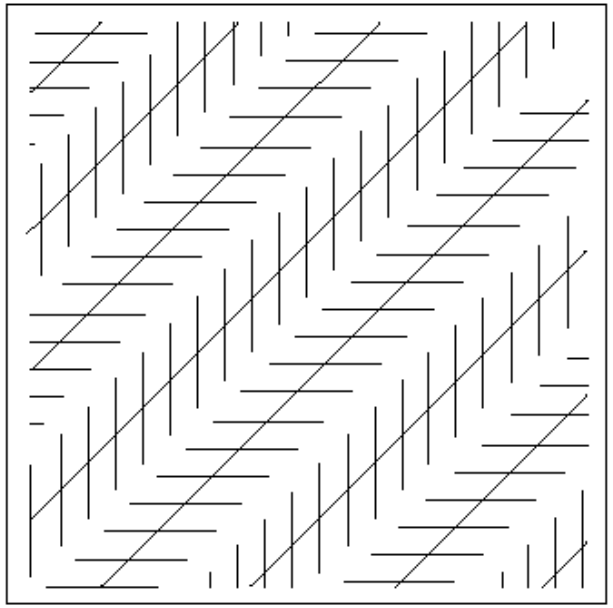
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- Brightness adaption
- Simultaneous contrast
- Mach band effect
- **Optical illusions**

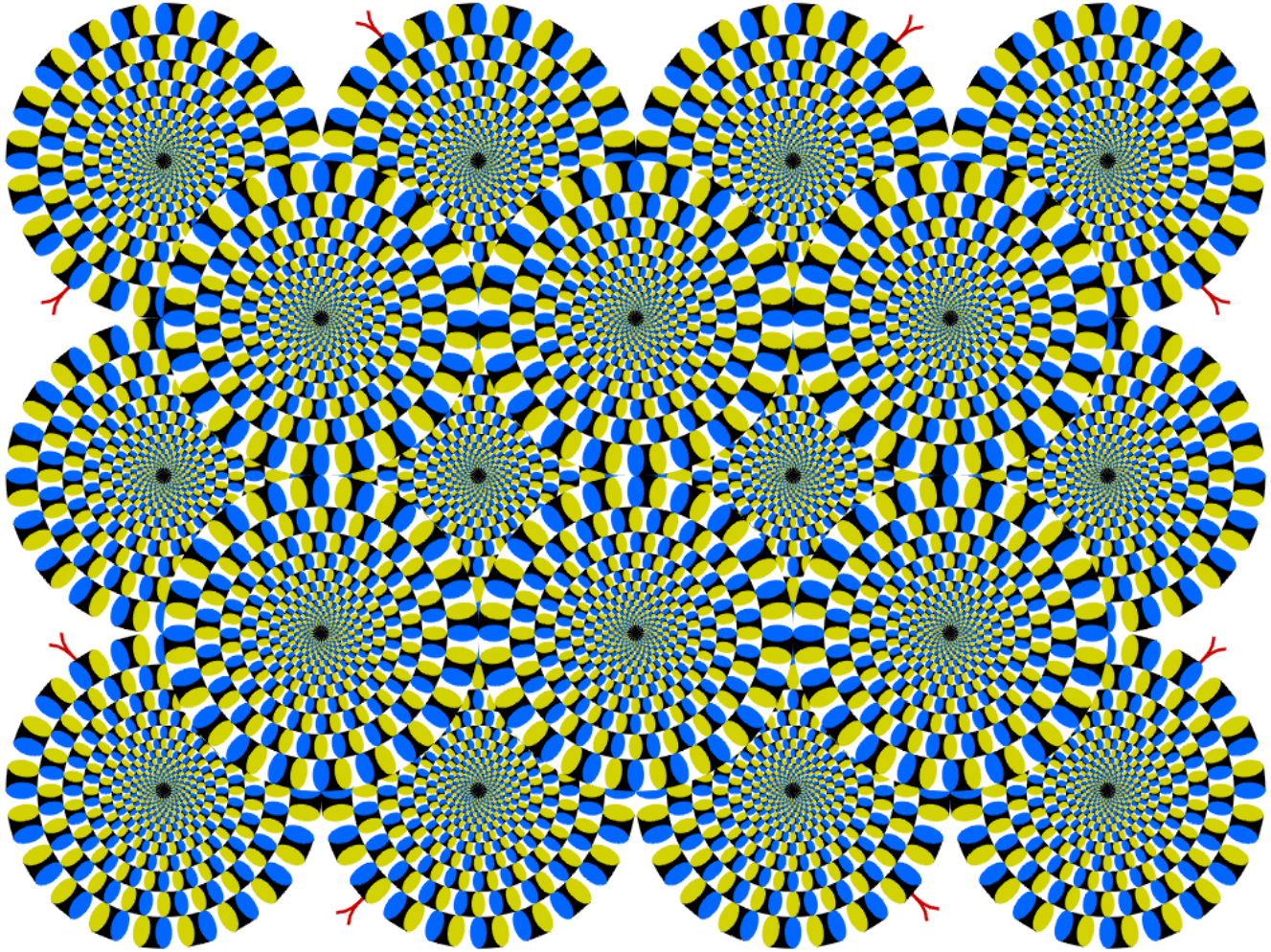




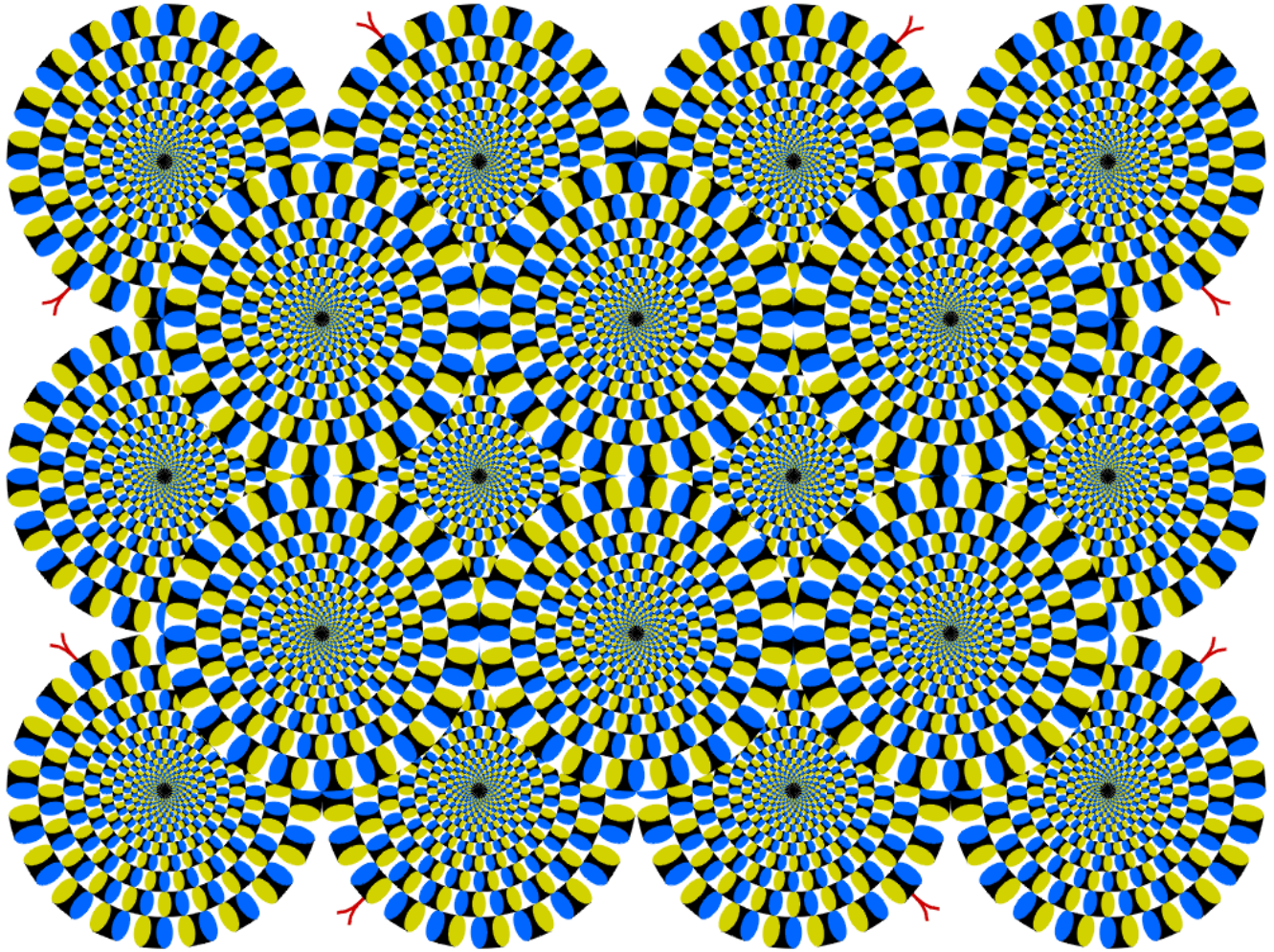




Characteristics of Human Visual System



Characteristics of Human Visual System





http://www.ritsumei.ac.jp/~akitaoka/index-e.html

Gender illusion





Your eyes can do more than you think...



- McGurk Effect

https://www.youtube.com/watch?v=G-lN8vWm3m0

https://www.youtube.com/watch?v=jtsfidRq2tw

Your eyes can do more than you think...



- Rubber hand illusion

https://www.youtube.com/watch?v=xdxlT68ygt8

https://www.youtube.com/watch?v=DphlhmtGRqI

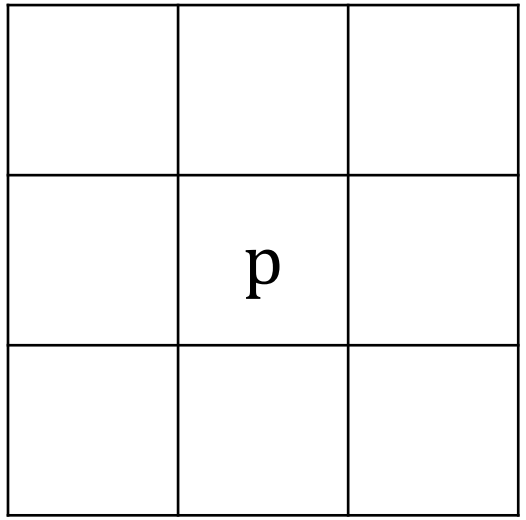
Pixel basics

- Neighbors of Pixel
- Relationship between Pixels
  - Adjacency
  - Connectivity
  - Regions
  - Boundaries
- Distance measures
  - Euclidean distance
  - City-block distance
  - Chessboard distance

Neighbors of Pixel

For a pixel p with coordinates (x, y)

- 4-*neighbors:* $N_4(p)$
- *diagonal neighbors:* $N_D(p)$
- *8-neighbors:* $N_8(p)$



Neighbors of Pixel

For a pixel p with coordinates (x, y)

- *4-neighbors:* $N_4(p)$

  (x+1, y), (x-1, y), (x, y+1), (x, y-1)

- *diagonal neighbors:* $N_D(p)$

- *8-neighbors:* $N_8(p)$

	q_1	
q_2	p	q_3
	q_4	

Neighbors of Pixel

For a pixel p with coordinates (x, y)

- *4-neighbors:* $N_4(p)$

  (x+1, y), (x-1, y), (x, y+1), (x, y-1)

- *diagonal neighbors:* $N_D(p)$

  (x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)

- *8-neighbors:* $N_8(p)$

r_1		r_2
	p	
r_3		r_4

Neighbors of Pixel

For a pixel p with coordinates (x, y)

- *4-neighbors:* **$N_4(p)$**

  (x+1, y), (x-1, y), (x, y+1), (x, y-1)

- *diagonal neighbors:* **$N_D(p)$**

  (x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)

- *8-neighbors:* **$N_8(p)$: $N_4(p) \cup N_D(p)$**

r_1	q_1	r_2
q_2	p	q_3
r_3	q_4	r_4

Neighborhood of Pixel

- Neighborhood of p:
  - Set of locations of the neighbors of p

- 2 types of neighborhoods:
  - Closed: contains p
  - Open: does not contain p

r_1	q_1	r_2
q_2	p	q_3
r_3	q_4	r_4

Adjacency

- **To define adjacency (邻接性) of pixels, we need identify**
- **Type of Neighbors**
  - $N_4(p)$, $N_D(p)$, $N_8(p)$
- **The set of intensity values V**
  - Binary image: V = {1}
  - Gray-scale image: V = [L_{min}, L_{max}]

q_1	p	q_2

0	0	0
1	1	1
0	0	0

Adjacency in a binary image

q_1	p	q_2

0	39	0
11	13	16
0	0	0

Adjacency in a gray-scale image

e.g. V = [10, 17], which pixels are adjacent?

Why adjacency? (E.g. Used for edge detection)

Adjacency

Types of Adjacency:

- 4-adjacency
- 8-adjacency
- M-adjacency (mixed adjacency)

Adjacency

Types of Adjacency:

- 4-adjacency
  - $p, q \in V$
  - $q \in N_4(p)$
- 8-adjacency
- M-adjacency (mixed adjacency)

r_{11}	r_{12}	r_{13}
r_{21}	r_{22}	r_{23}
r_{31}	r_{32}	r_{33}

0	1 ·····	1
	⋮	
0	1	0
0	0	1

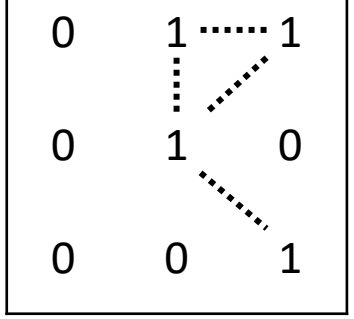
V = {1}

Adjacency

Types of Adjacency:

- 4-adjacency
- 8-adjacency
  - $p, q \in V$
  - $q \in N_8(p)$
- M-adjacency (mixed adjacency)

r_{11}	r_{12}	r_{13}
r_{21}	r_{22}	r_{23}
r_{31}	r_{32}	r_{33}



V = {1}

Adjacency

Types of Adjacency:

- 4-adjacency
- 8-adjacency
- M-adjacency (mixed adjacency)
  - $p, q \in V$
  - $q \in N_4(p)$ or
  - $q \in N_D(p)$ and $N_4(p) \cap N_4(q) \notin V$

r_{11}	r_{12}	r_{13}
r_{21}	r_{22}	r_{23}
r_{31}	r_{32}	r_{33}

0	1	1
0	1	0
0	0	1

?

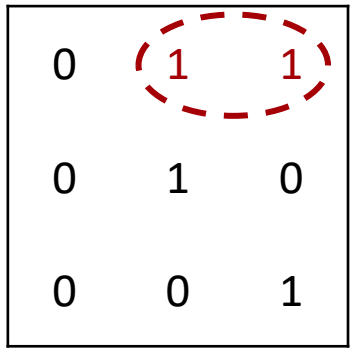
Adjacency

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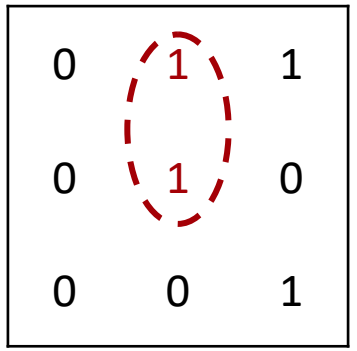
Mixed adjacency:

- $p, q \in V$
- $q \in N_4(p)$ or
- $q \in N_D(p)$ and $N_4(p) \cap N_4(q) \notin V$

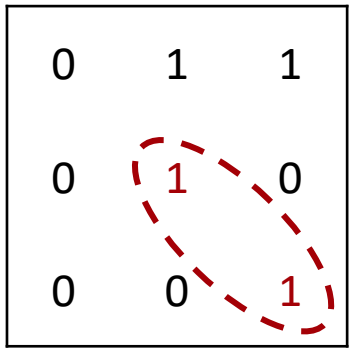
$V = \{1\}$



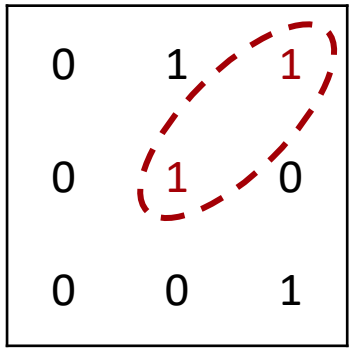
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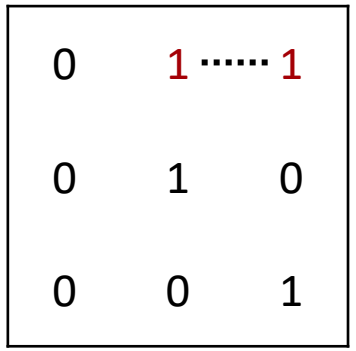
Adjacency

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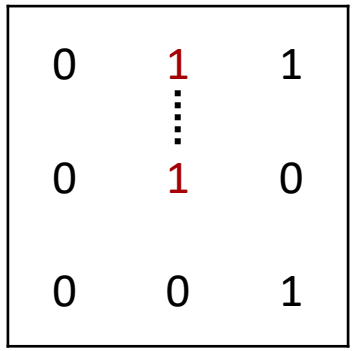
Mixed adjacency:

- $p, q \in V$
- $q \in N_4(p)$ or
- $q \in N_D(p)$ and $N_4(p) \cap N_4(q) \notin V$

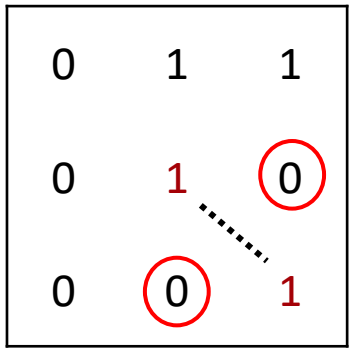
$V = \{1\}$



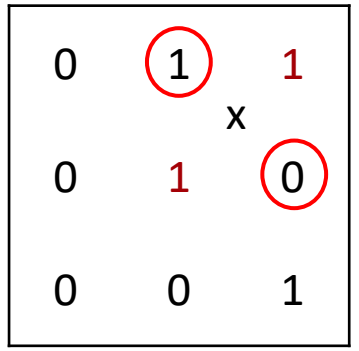
$q \in N_4(p)$



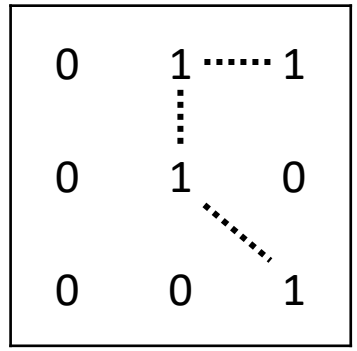
$q \in N_4(p)$



$q \in N_D(p)$
$N_4(p) \cap N_4(q) \notin V$



$q \in N_D(p)$
~~$N_4(p) \cap N_4(q) \notin V$~~



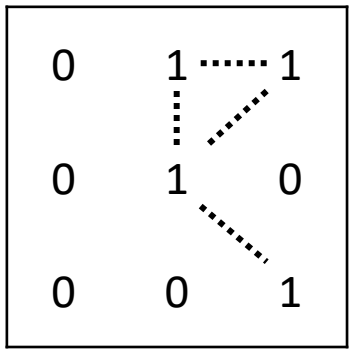
M-adjacencies

M-adjacency is a modification of 8-adjacency, to eliminate the ambiguities that may result from using 8-adjacency when defining a path between points

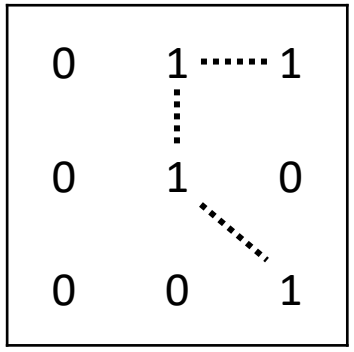
Digital Path (or curve)

- Digital path from pixel $p(x_0, y_0)$ to $q(x_n, y_n)$
  - Sequence $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$
  - Where (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for$1 \leq i \leq n$
- If $(x_0, y_0) = (x_n, y_n)$: closed path
- n is the length of the path
- Depends on the definition of adjacency used

r_{11}	r_{12}	r_{13}
r_{21}	r_{22}	r_{23}
r_{31}	r_{32}	r_{33}



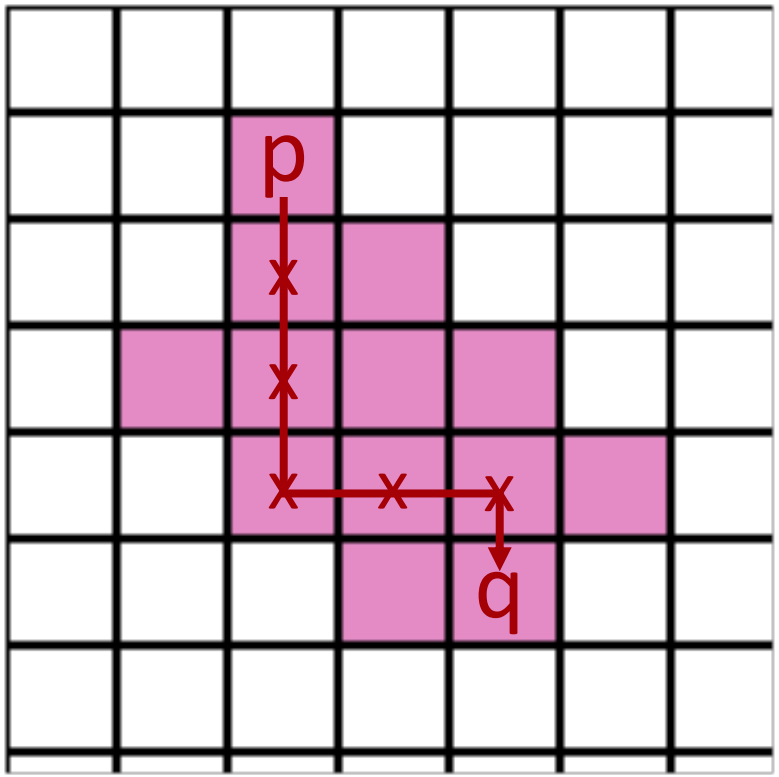
8-paths

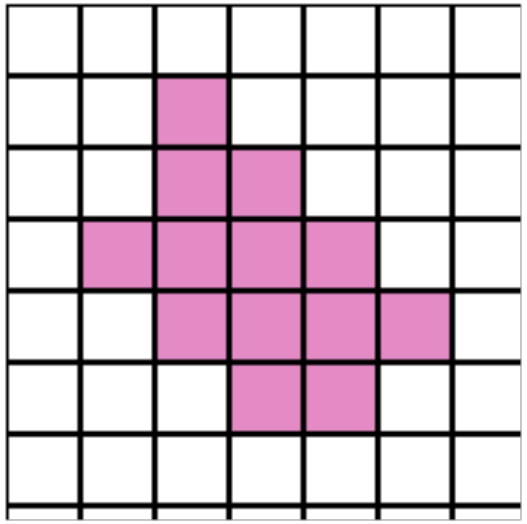


m-path

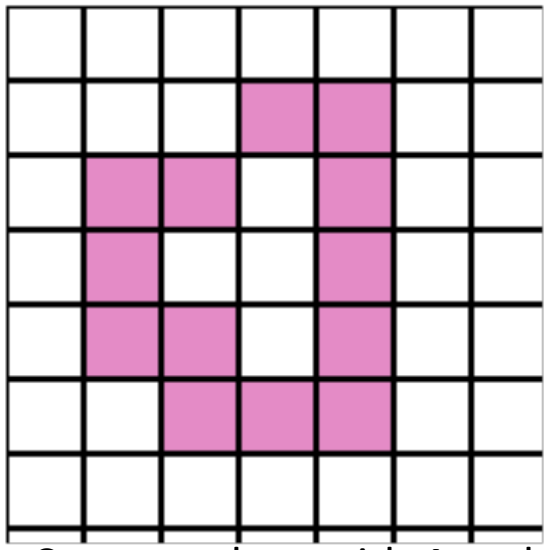
Connectivity

- Pixels p and q are connected in S if there exists a path between them consisting entirely of pixels in S.
- For any pixel p in S, the set of pixels that are connected to it in S is called a connected component of S
- If it only has one component, and that component is connected, then S is called a connected set.

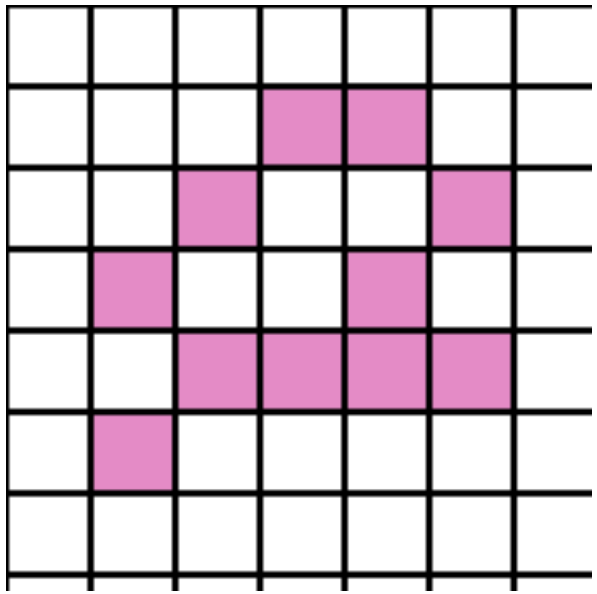




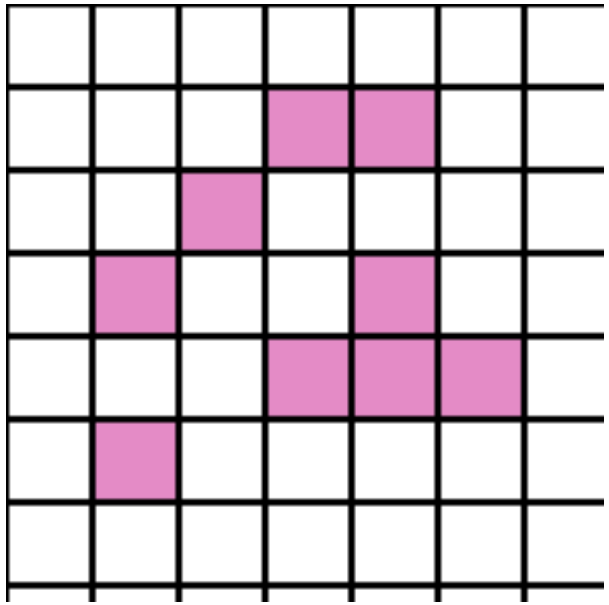
Connected set with 4-path



Connected set with 4-path



NOT Connected set with 4-path
Connected set with 8-path



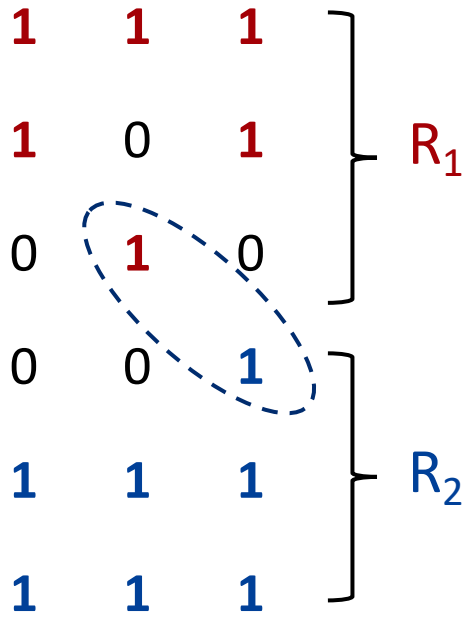
NOT a Connected set
3 connected components with 8-path

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Region

- ***R***: a subset of an image which is also a connected set

- Adjacent region: their union is a connected set
  - Disjoint region otherwise

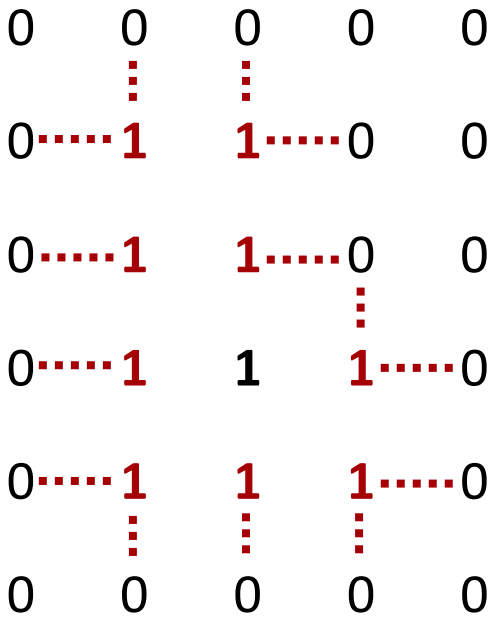


Boundary

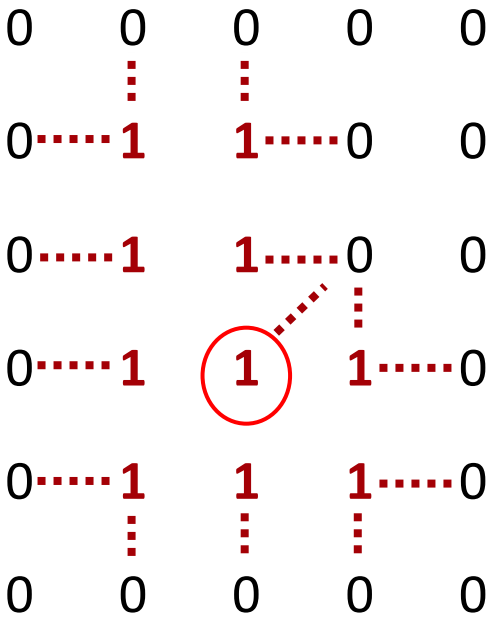
- A set of pixels of R that are adjacent to pixels in the complement of R.
  - Inner border and outer border
  - Image border

0	0	0	0	0
0	**1**	**1**	0	0
0	**1**	**1**	0	0
0	**1**	**1**	**1**	0
0	**1**	**1**	**1**	0
0	0	0	0	0

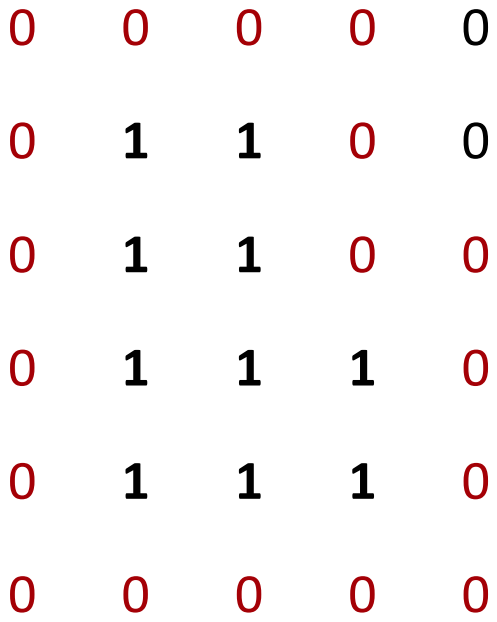
Boundary



Inner boundary:
4-adjacency



Inner boundary:
8-adjacency



Outer boundary
8-adjacency

Note: Pixels on image border are included in boundaries !

Boundary VS Edge

Boundary

- Closed path
- “global” concept

Edge

- Defined at pixel level (for instance: derivatives)
- Local concept
- Can be linked into boundaries



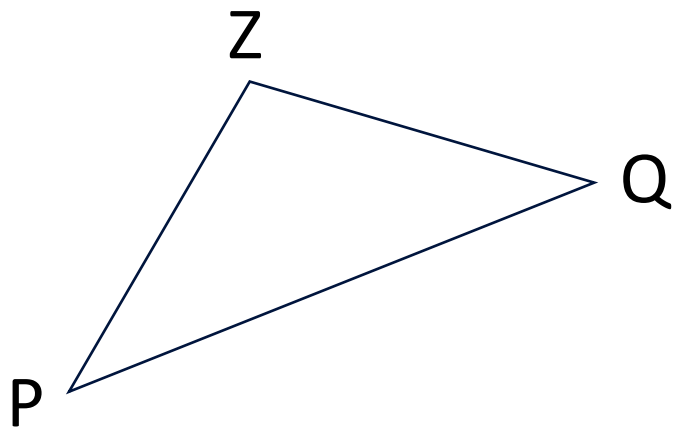
Pixel basics

- Neighbors of Pixel
- Relationship between Pixels
  - Adjacency
  - Connectivity
  - Regions
  - Boundaries
- **Distance measures**
  - Euclidean distance
  - City-block distance
  - Chessboard distance

Distance Measures

- For pixels p, q and z, with coordinates (x, y), (s, t) and (v, w), D is a distance function or metric if

  - $D(p, q) \geq 0$ (with $D(p, q) = 0$ if and only if $p = q$)
  - $D(p, q) = D(q, p)$
    commutative
  - $D(p, q) \leq D(p, z) + D(q, z)$
    triangle inequality



Distance Measures

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$p(x, y)$
$q(s, t)$

- **Euclidean distance:**

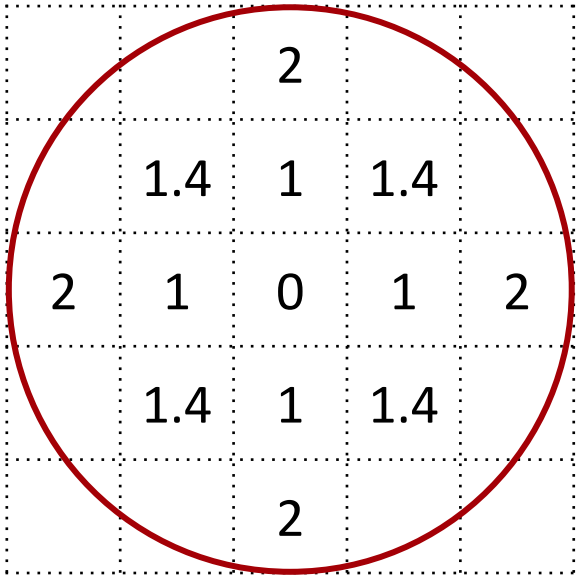
  $D_e(p, q) = [(x-s)^2+(y-t)^2]^{1/2}$

- City-block distance:

  $D_4(p, q) = |x-s|+|y-t|$

- Chessboard distance

  $D_8(p, q) = \max(|x-s|, |y-t|)$



Distance Measures

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- Euclidean distance:

$$D_e(p, q) = [(x-s)^2+(y-t)^2]^{1/2}$$

- **City-block distance:**

$$D_4(p, q) = |x-s|+|y-t|$$

- Chessboard distance

$$D_8(p, q) = \max(|x-s|, |y-t|)$$

$p(x, y)$
$q(s, t)$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

The pixels with D_4=1 are the 4-neighbors of (x,y)

Distance Measures

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- **Euclidean distance:**

$$D_e(p, q) = [(x-s)^2+(y-t)^2]^{1/2}$$

- **City-block distance:**

$$D_4(p, q) = |x-s|+|y-t|$$

- **Chessboard distance**

$$D_8(p, q) = \max(|x-s|, |y-t|)$$

$p(x, y)$
$q(s, t)$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

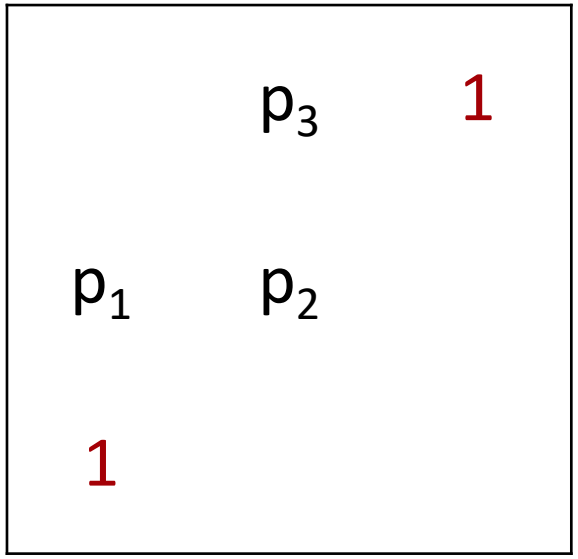
The pixels with D_8=1 are the 8-neighbors of (x,y)



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Distance Measures

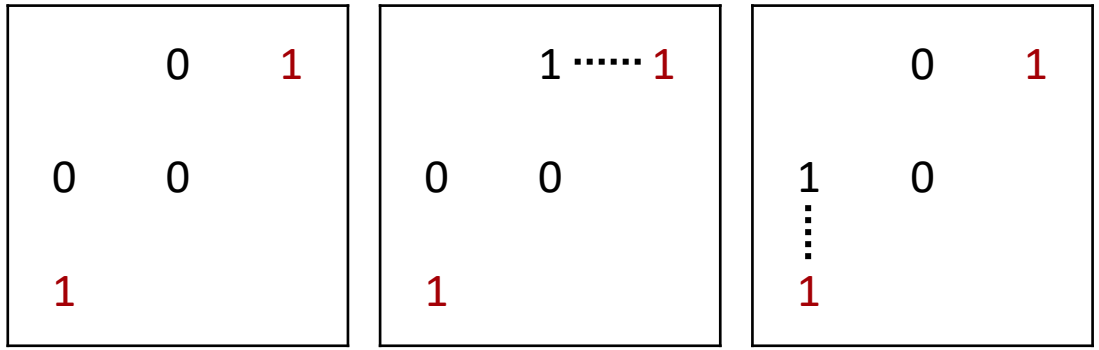
- D_m distance is defined as the shortest m-path (i.e. with mixed-adjacency) between the points



$$\mathbf{D_m = ?}$$

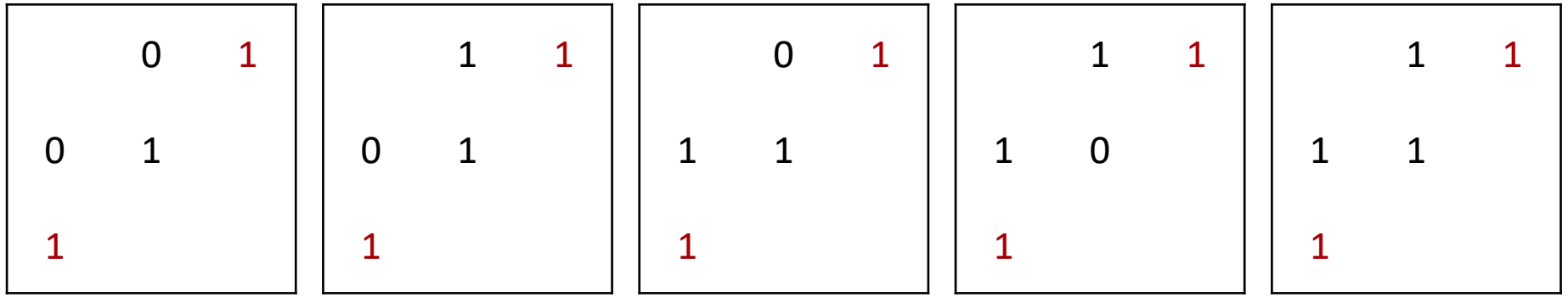
Distance Measures

- **No m-path between the point**



Distance Measures

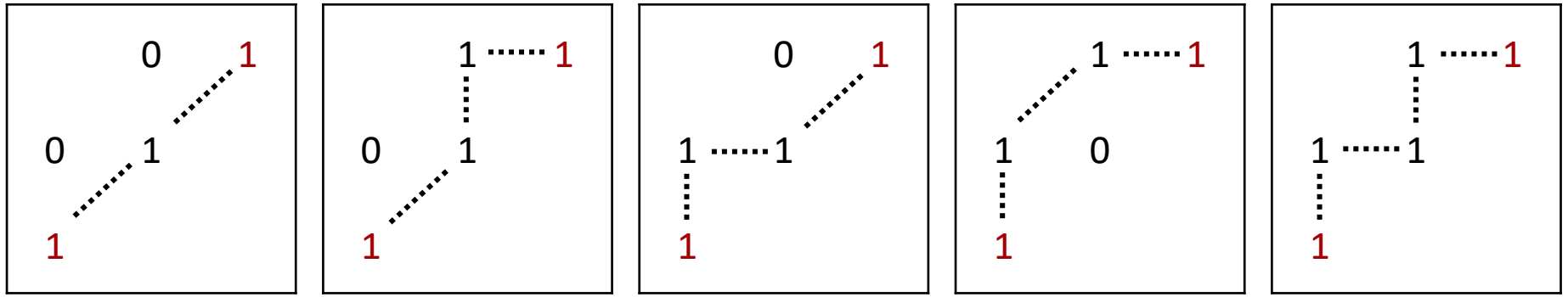
- **D_m distance** is different by the values of p_1, p_2 and p_3



D_m = ?　　D_m = ?　　D_m = ?　　D_m = ?　　D_m = ?

Distance Measures

- **D_m distance** is different by the values of p_1, p_2 and p_3



$D_m = 2$ $D_m = 3$ $D_m = 3$ $D_m = 3$ $D_m = 4$

Arithmetic Operations between images

- Addition

$$s(x,y) = f(x,y) + g(x,y)$$

- Subtraction

$$d(x,y) = f(x,y) - g(x,y)$$

- Multiplication

$$p(x,y) = f(x,y) \times g(x,y)$$

- Division

$$v(x,y) = f(x,y) \div g(x,y)$$

Array and Matrix Operation

- Consider two 2x2 images

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

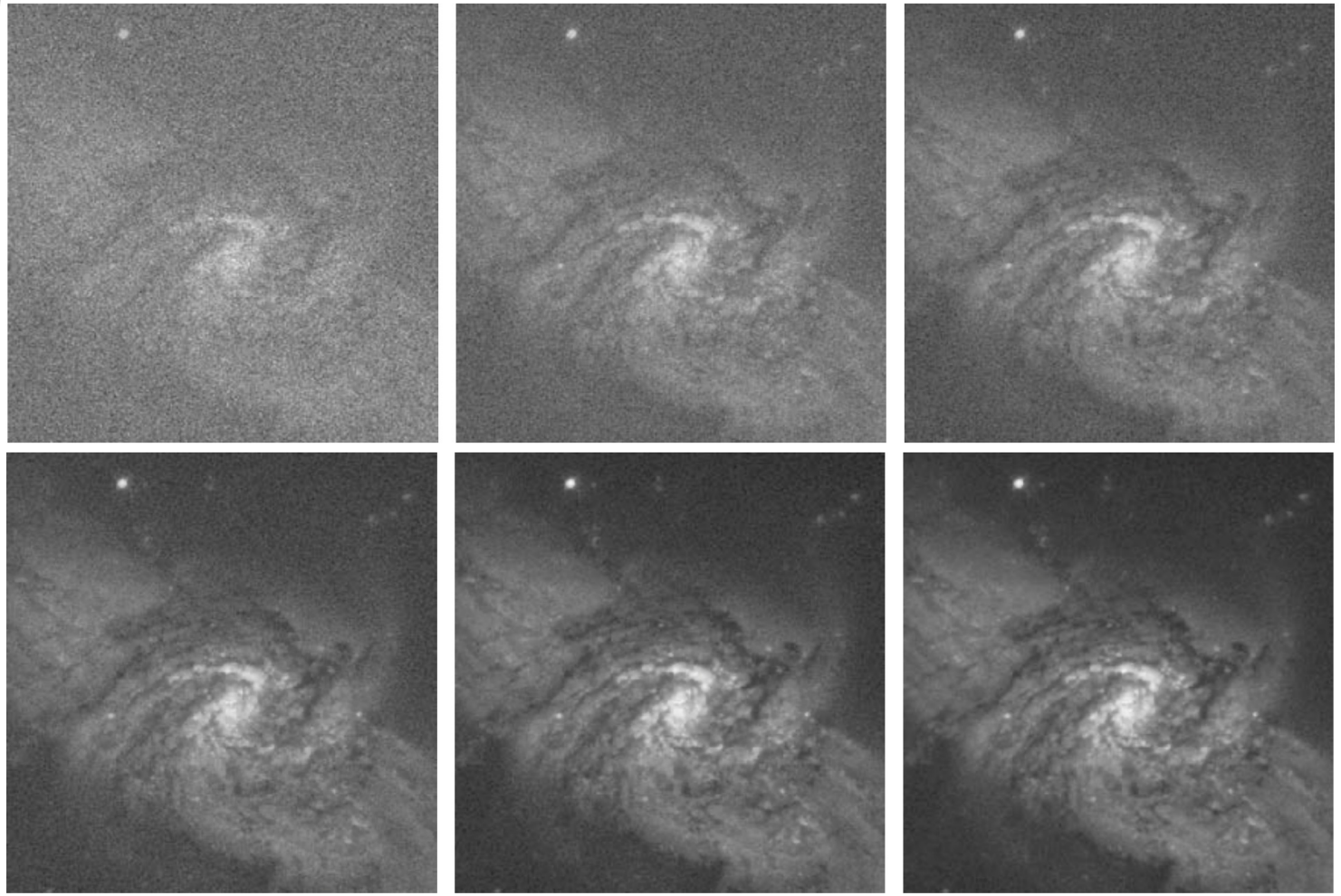
- Matrix product (General matrix product: 一般矩阵乘积)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

- Array product $\odot$ or $\otimes$ (Elementwise product, Hadamard Product: 哈达玛积)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Image Addition



a b c
d e f

FIGURE 2.29 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)-(f) Result of averaging 5, 10, 20, 50, and 1,00 noisy images, respectively. All images are of size 566×598 pixels, and all were scaled so that their intensities would span the full [0, 255] intensity scale. (Original image courtesy of NASA.)

Image Addition – Night Mode



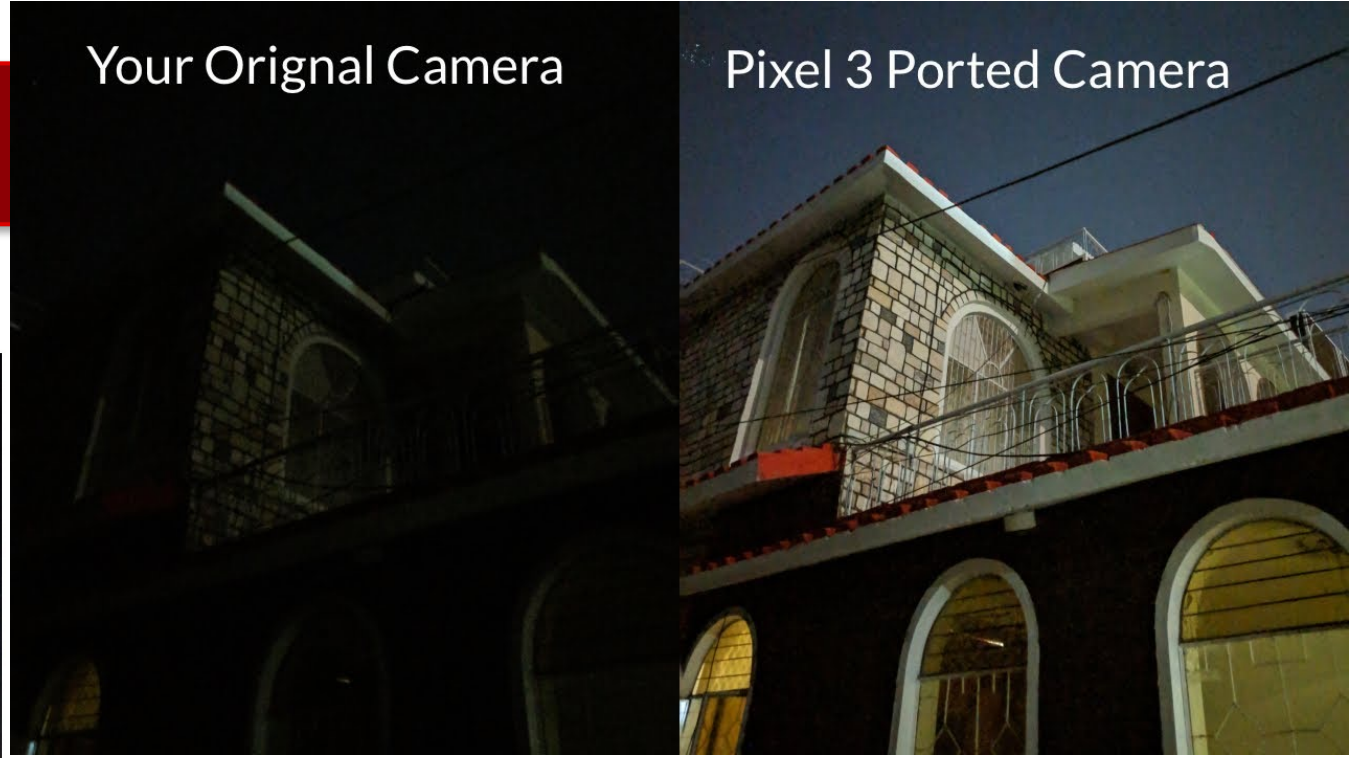
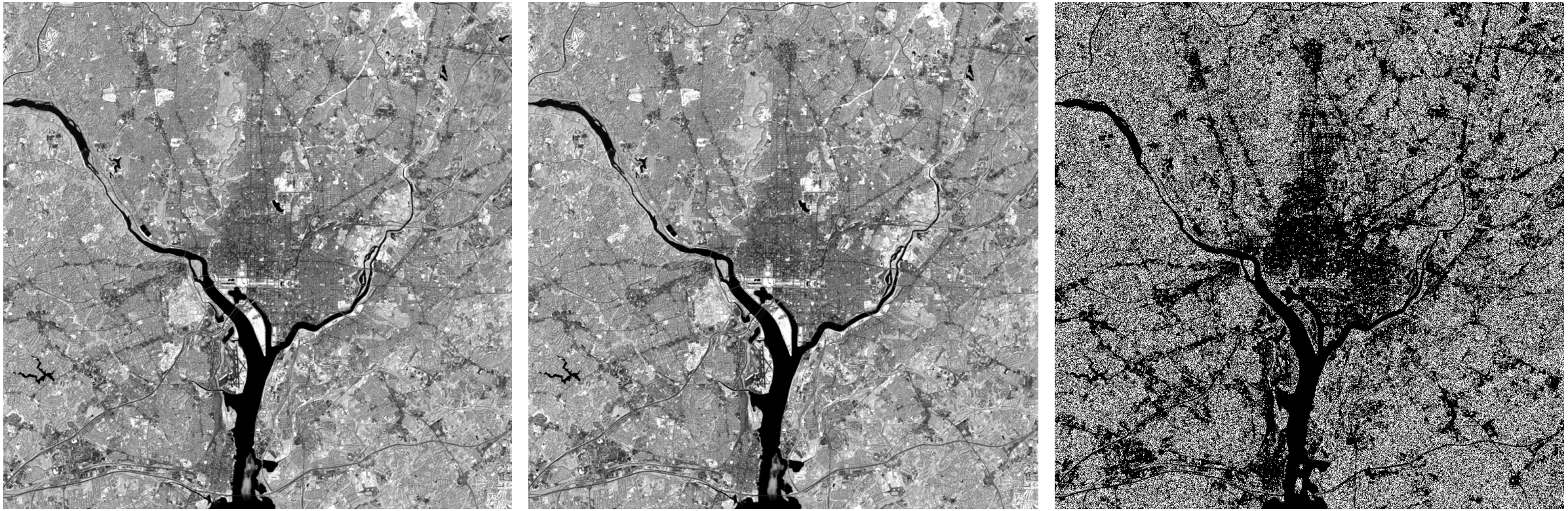






Image Subtraction



a b c

FIGURE 2.30 (a) Infrared image of the Washington, D.C. area. (b) Image resulting from setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range [0, 255] for clarity. (Original image courtesy of NASA.)

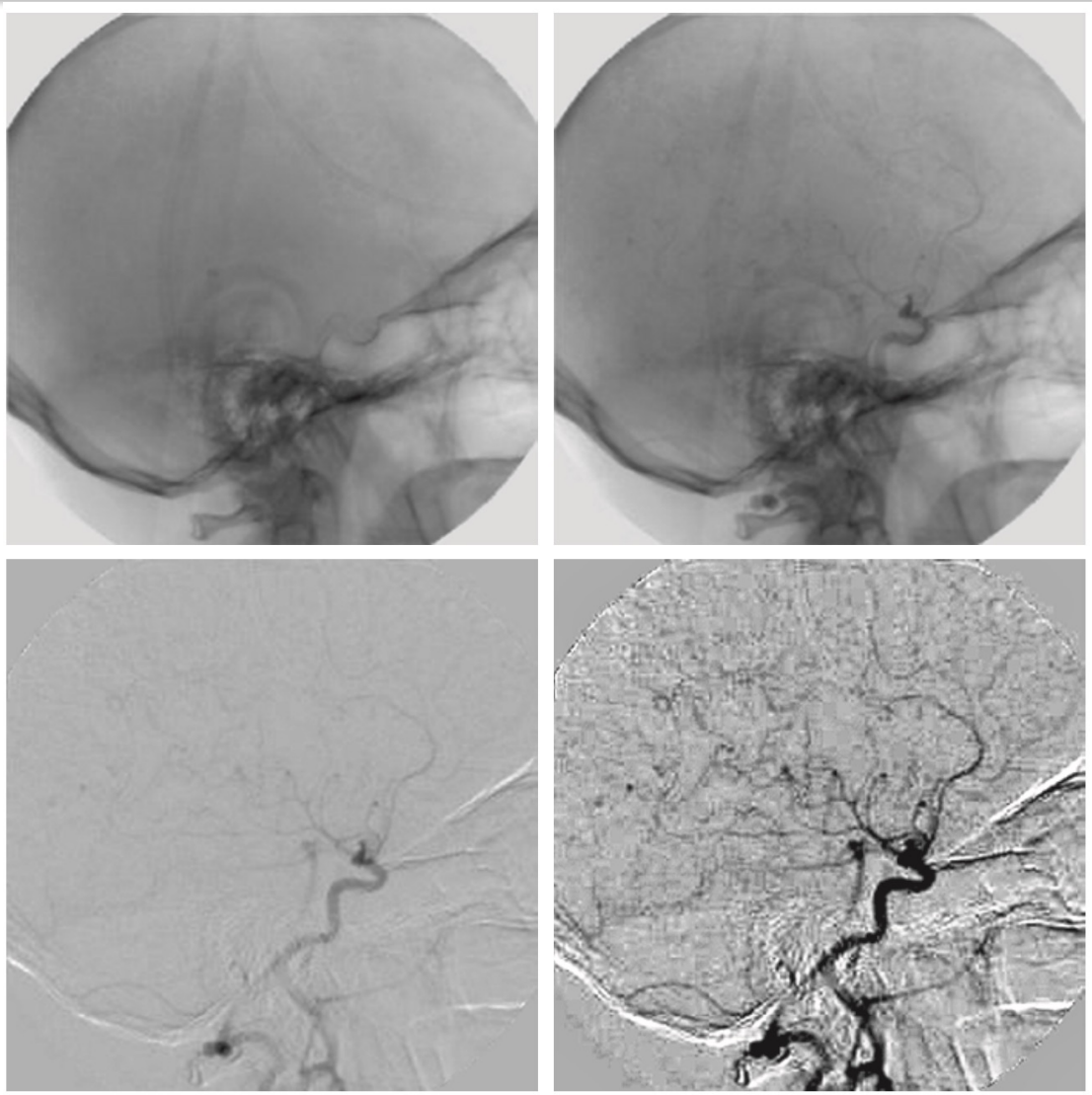
Image Subtraction



a b c

FIGURE 2.31 (a) Difference between the 930 dpi and 72 dpi images in Fig. 2.23. (b) Difference between the 930 dpi and 150 dpi images. (c) Difference between the 930 dpi and 300 dpi images.

Image Subtraction

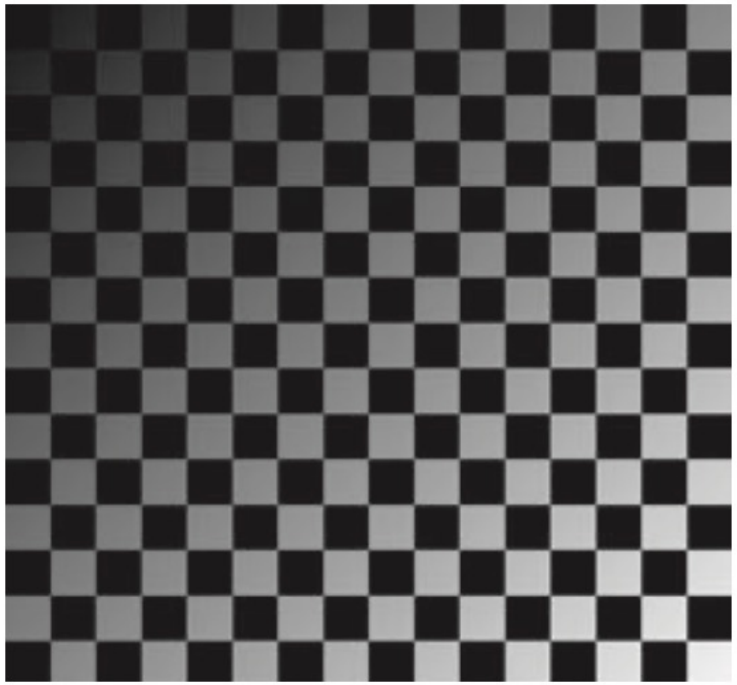


a b
c d

FIGURE 2.32 Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of the Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)

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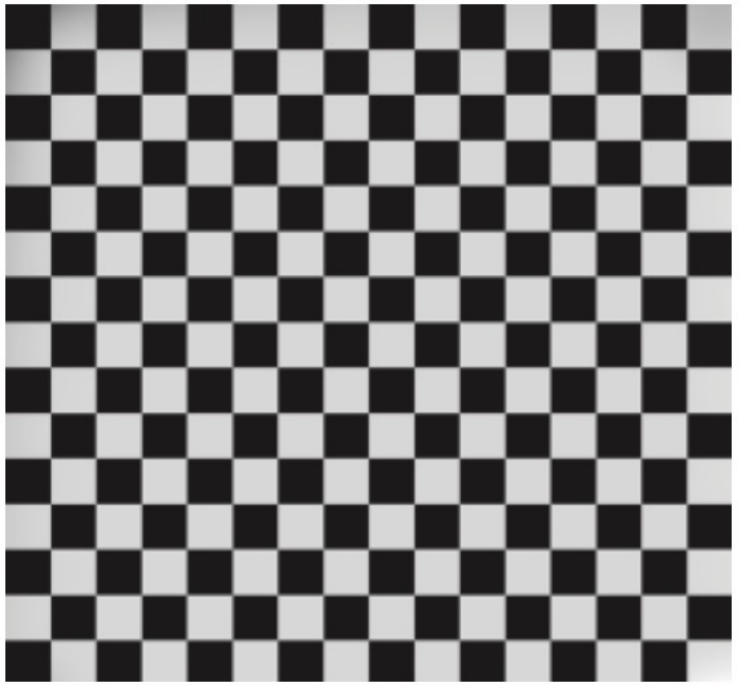
Image Division



$g(x,y) = f(x,y) \times h(x,y)$



$h(x,y)$



$f(x,y)$

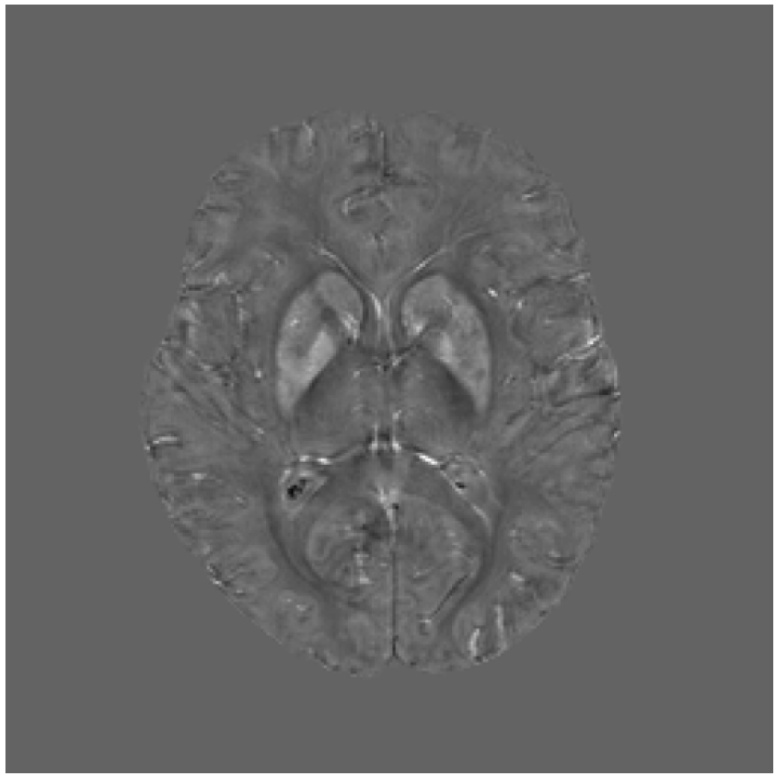
$$f(x,y) = g(x,y)/h(x,y)$$

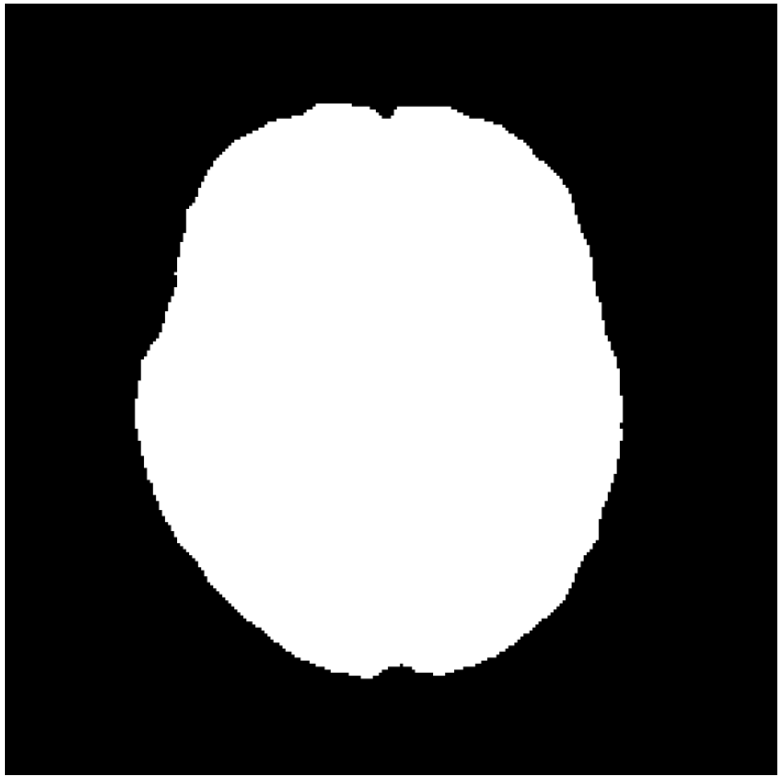
Image Multiplication : Region Of Interest (ROI)

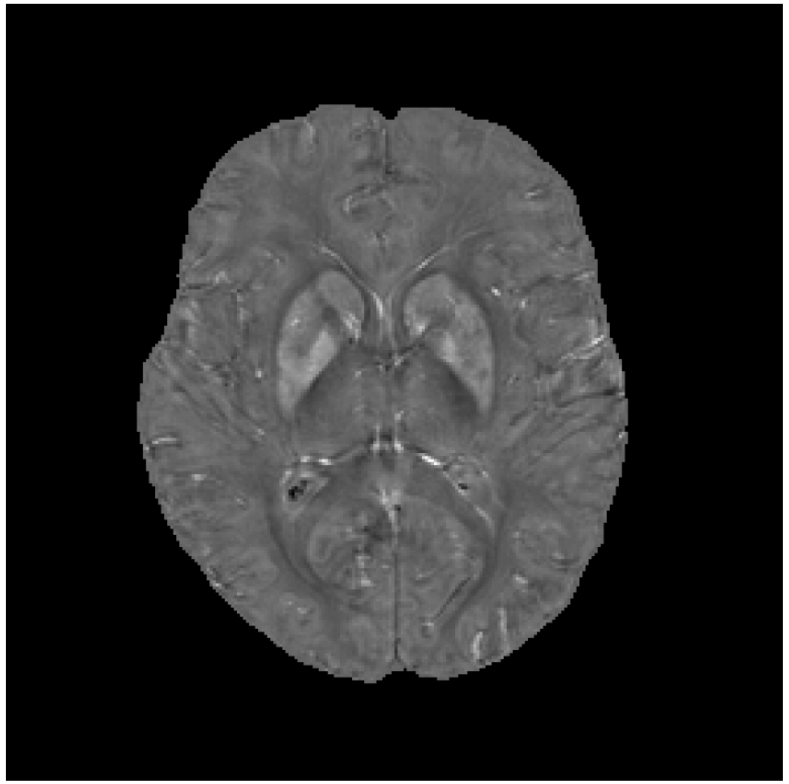


$$p(x,y) = f(x,y) \times g(x,y)$$

Background removal (masking)



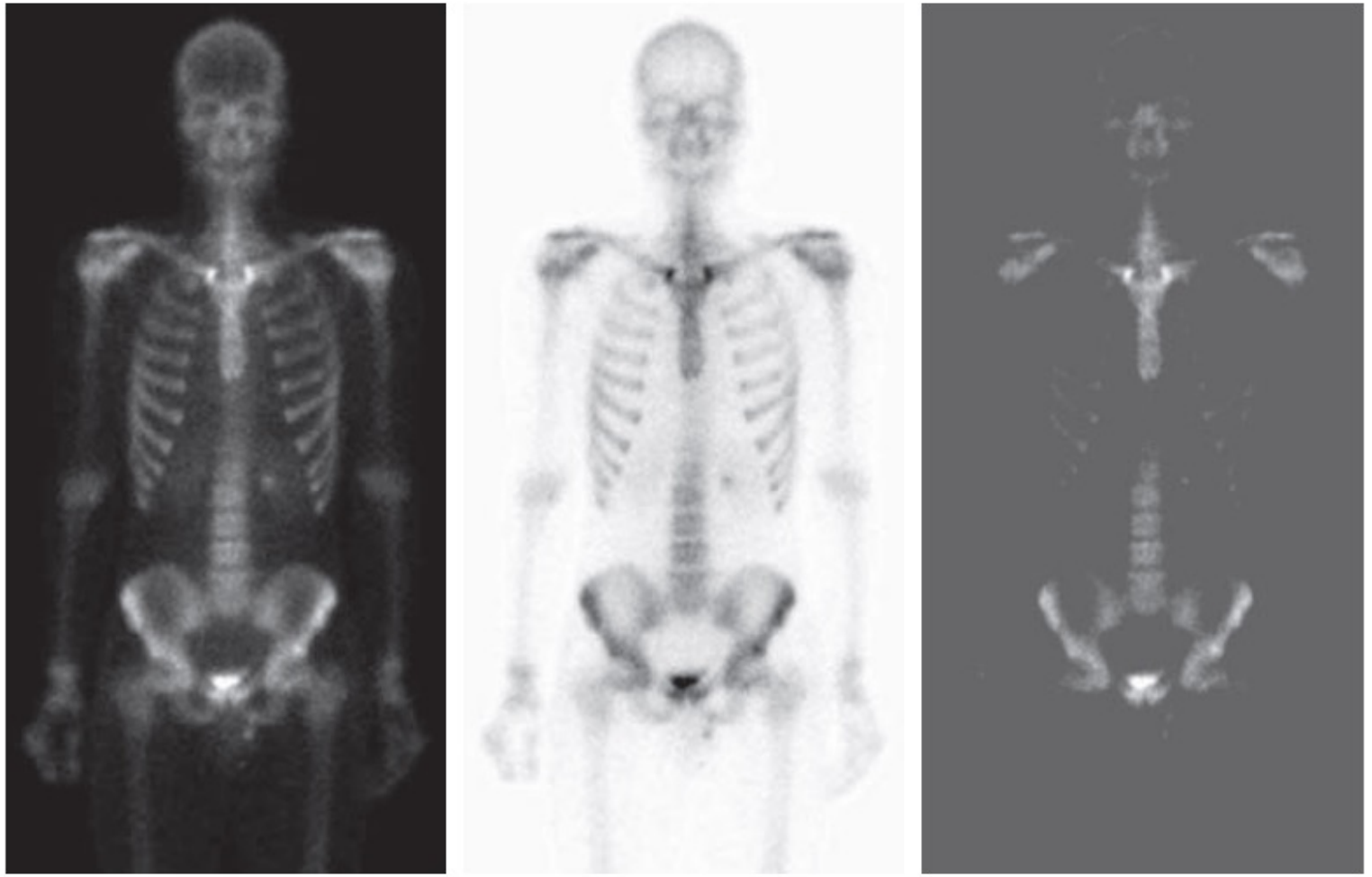




The negative

a b c

FIGURE 2.36
Set operations involving grayscale images. (a) Original image. (b) Image negative obtained using grayscale set complementation. (c) The union of image (a) and a constant image. (Original image courtesy of G.E. Medical Systems.)



Geometric transformation

- Modify the spatial arrangement of pixels in an image
- *Rubber-sheet transformations*
- To transform an image, two basic operations:
  - Spatial transformation of coordinates
  - Intensity interpolation to fill “gaps”

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

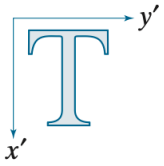
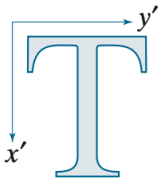
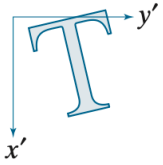
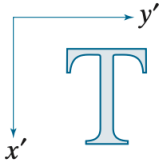
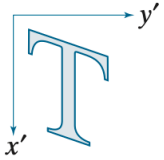
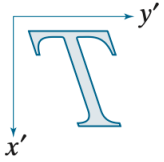
Transformed　　original

Affine transformations

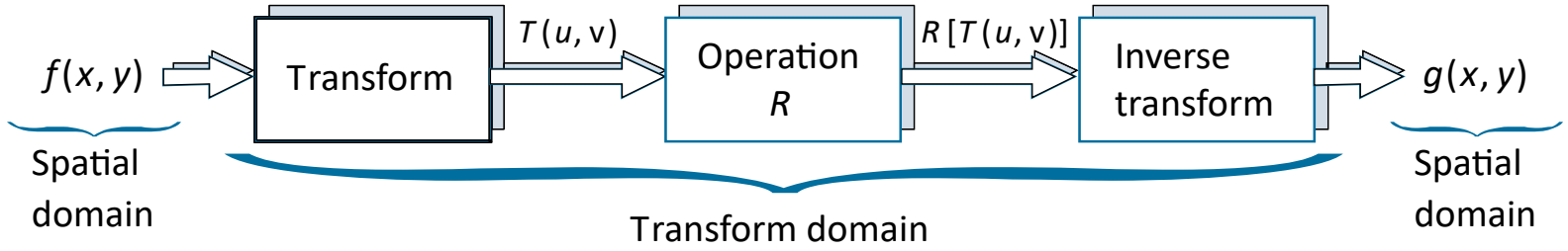
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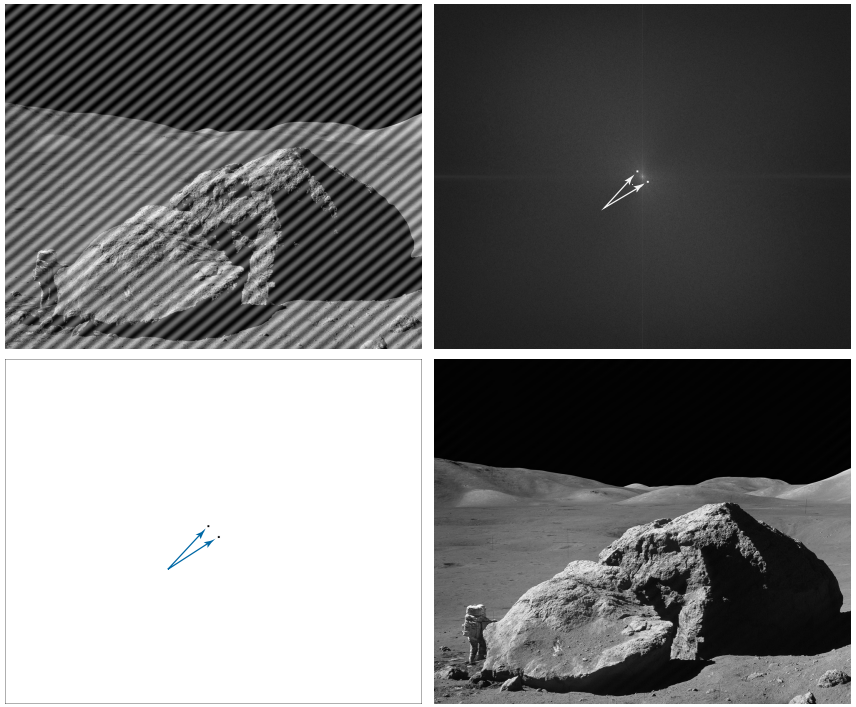
- Scaling, translation, rotation, shearing
- Preserves points, straight lines, planes
- Use homogenous coordinates to generalize transform (add translation)
- Can concatenate operations by multiplying their transform matrices

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to –1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x\cos\theta - y\sin\theta$ $y' = x\sin\theta + y\cos\theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	

Transform domain





a b
c d

FIGURE 2.45
(a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy caused by the interference (the bursts were enlarged for display purposes). (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)